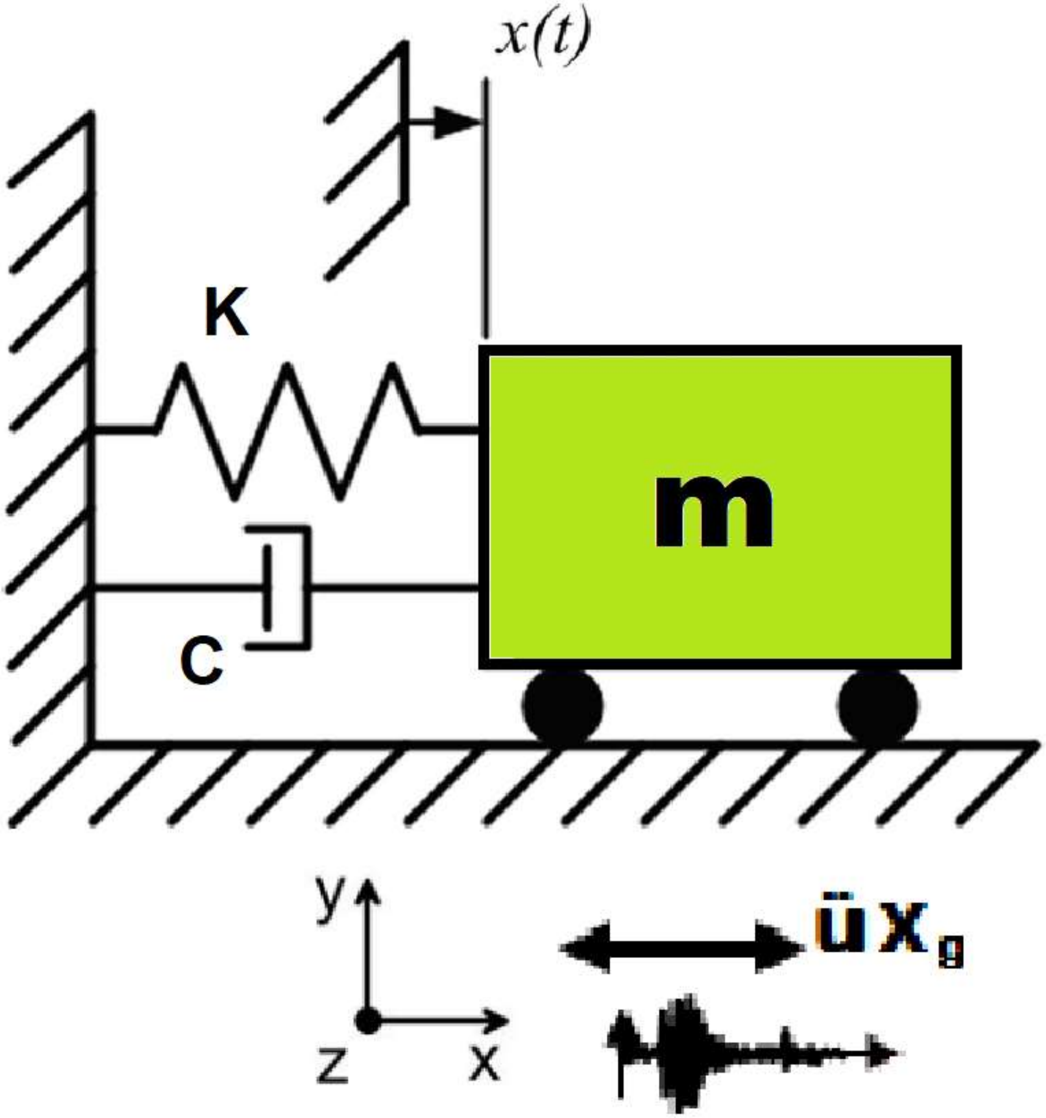
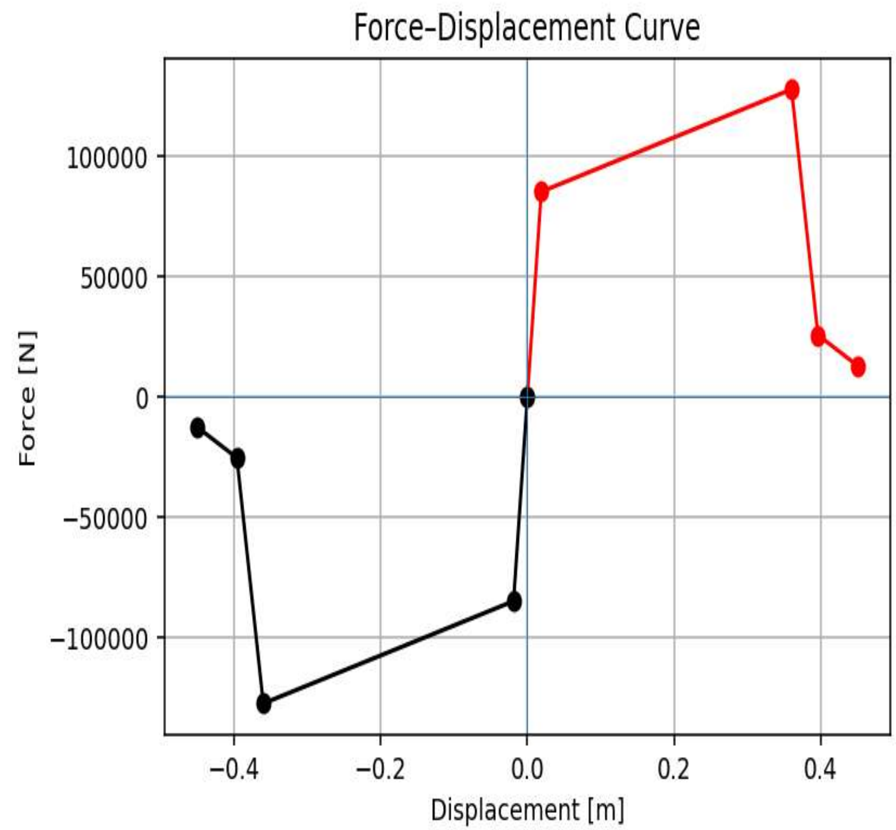
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

NONLINEAR DYNAMIC ANALYSIS UNDER A SINGLE GROUND MOTION RECORD COMPUTATION AND VISUALIZATION RESPONSE SPECTRA OF ACCELERATION, VELOCITY, DISPLACEMENT DUCTILITY DAMAGE INDEX USING OPENSEES

(CONSTANT STRUCTURAL DUCTILITY RATIO RESPONSE SPECTRUM)

WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)


x(t)
K
m
C
y
z
x
üx g

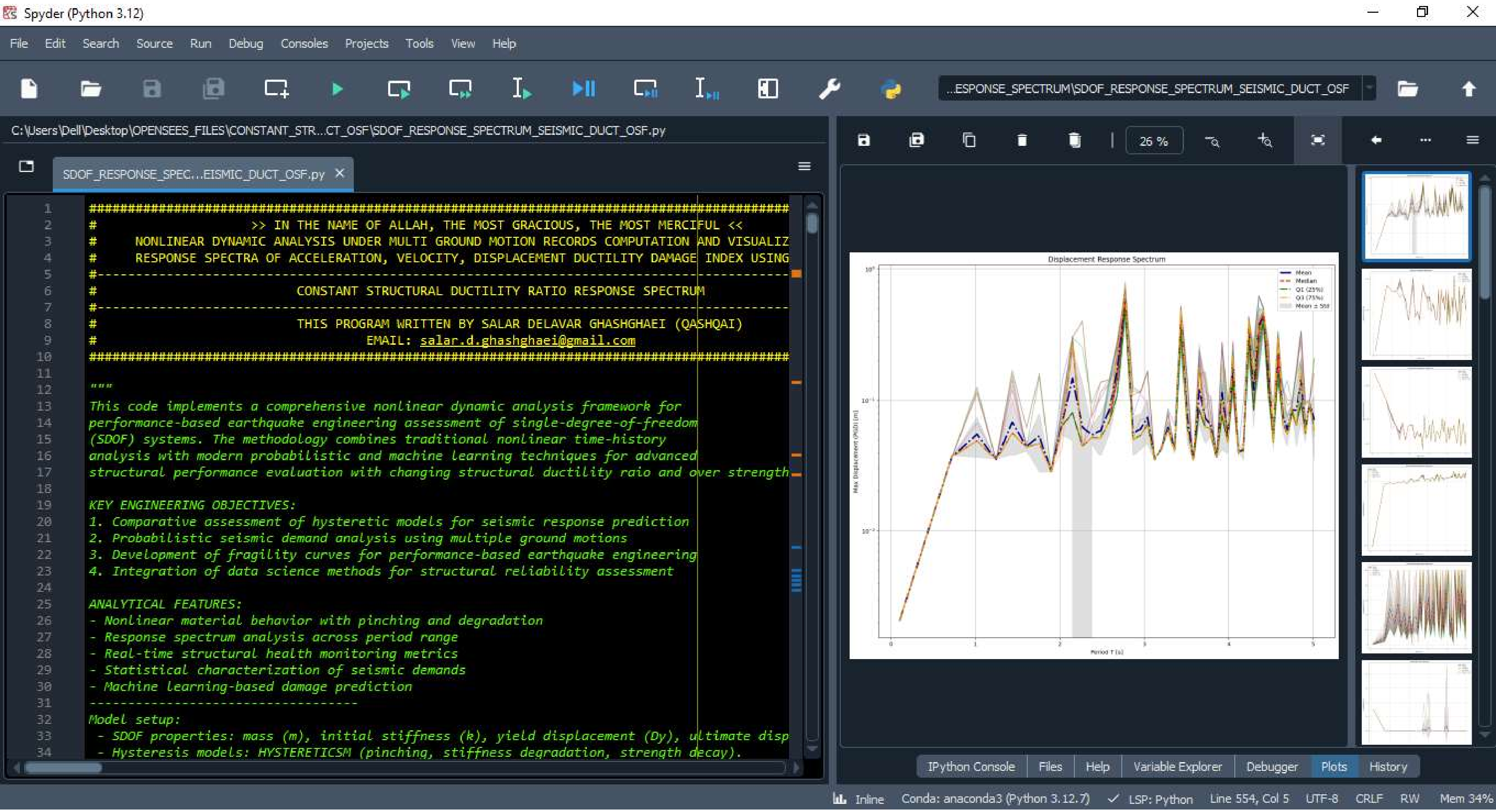

Force-Displacement Curve
100000
50000
0
−50000
−100000
Force [N]
−0.4
−0.2
0.0
0.2
0.4
Displacement [m]

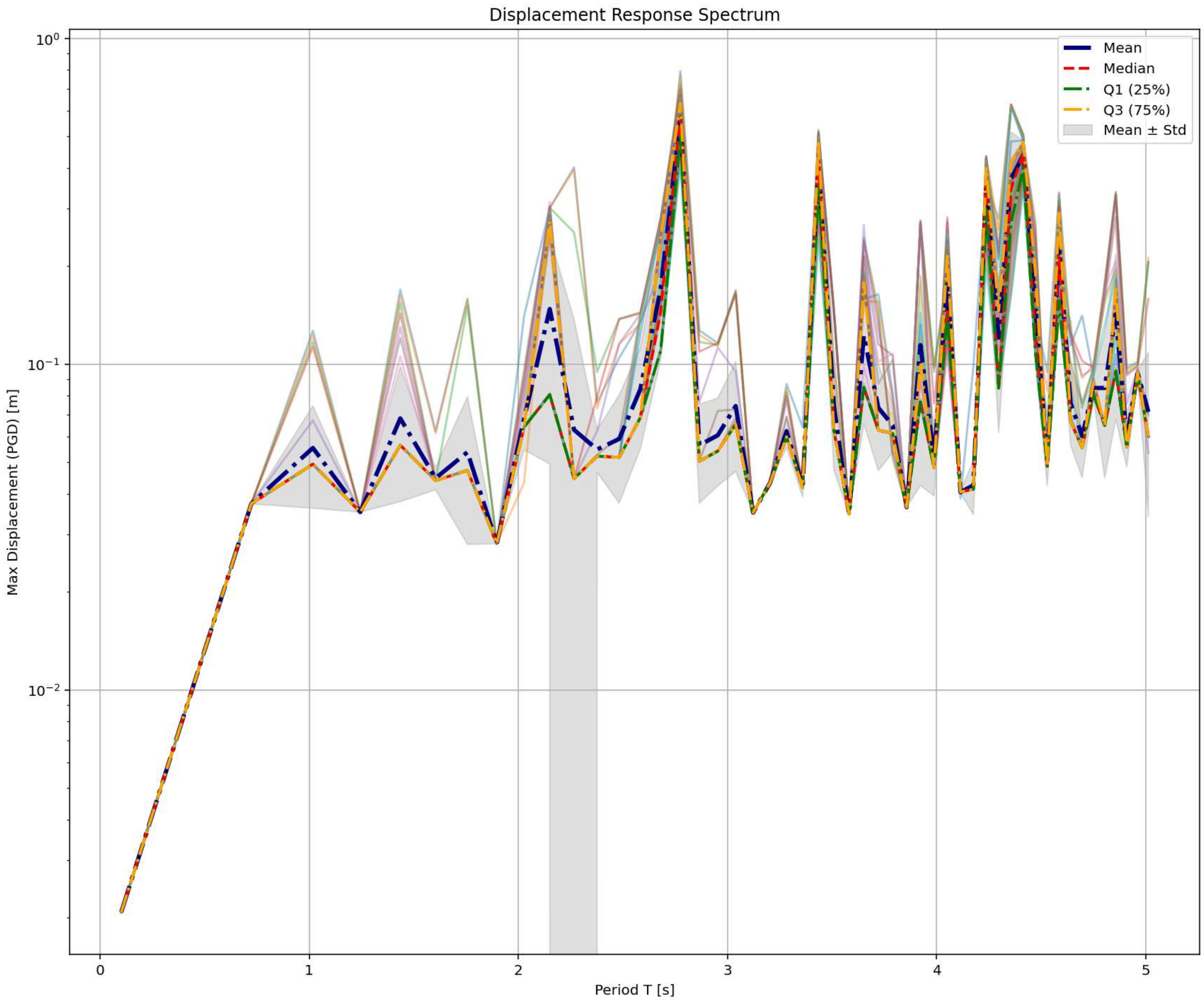
$$Structural\ Ductility\ Damage\ Index = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

$$\Delta_d = Lateral\ Displaement\ from\ Dynamic\ Analysis$$

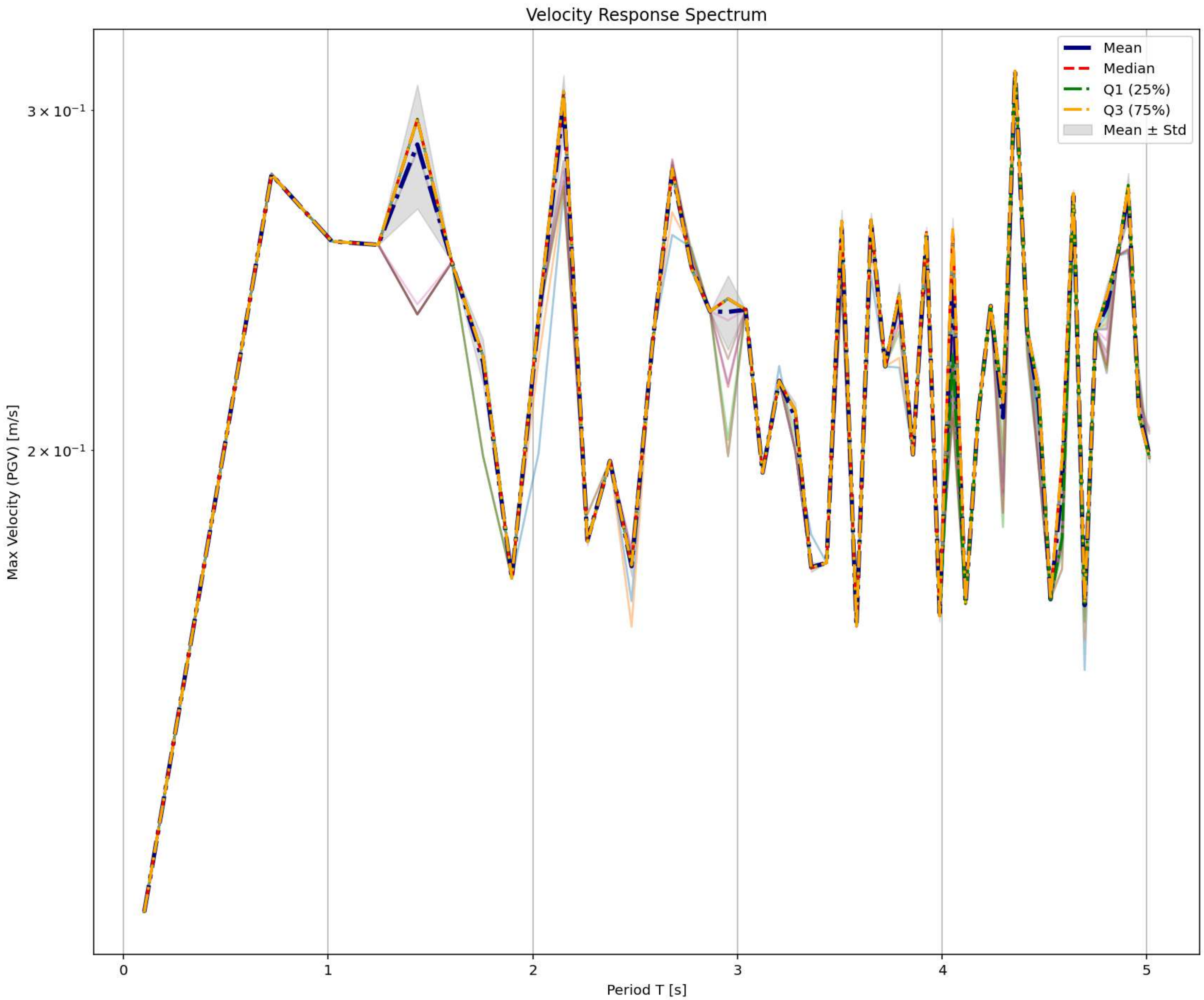
$$\Delta_y = Lateral\ Yield\ Displaement\ from\ Pushover\ Analysis$$

$$\Delta_u = Lateral\ Ultimate\ Displaement\ from\ Pushover\ Analysis$$

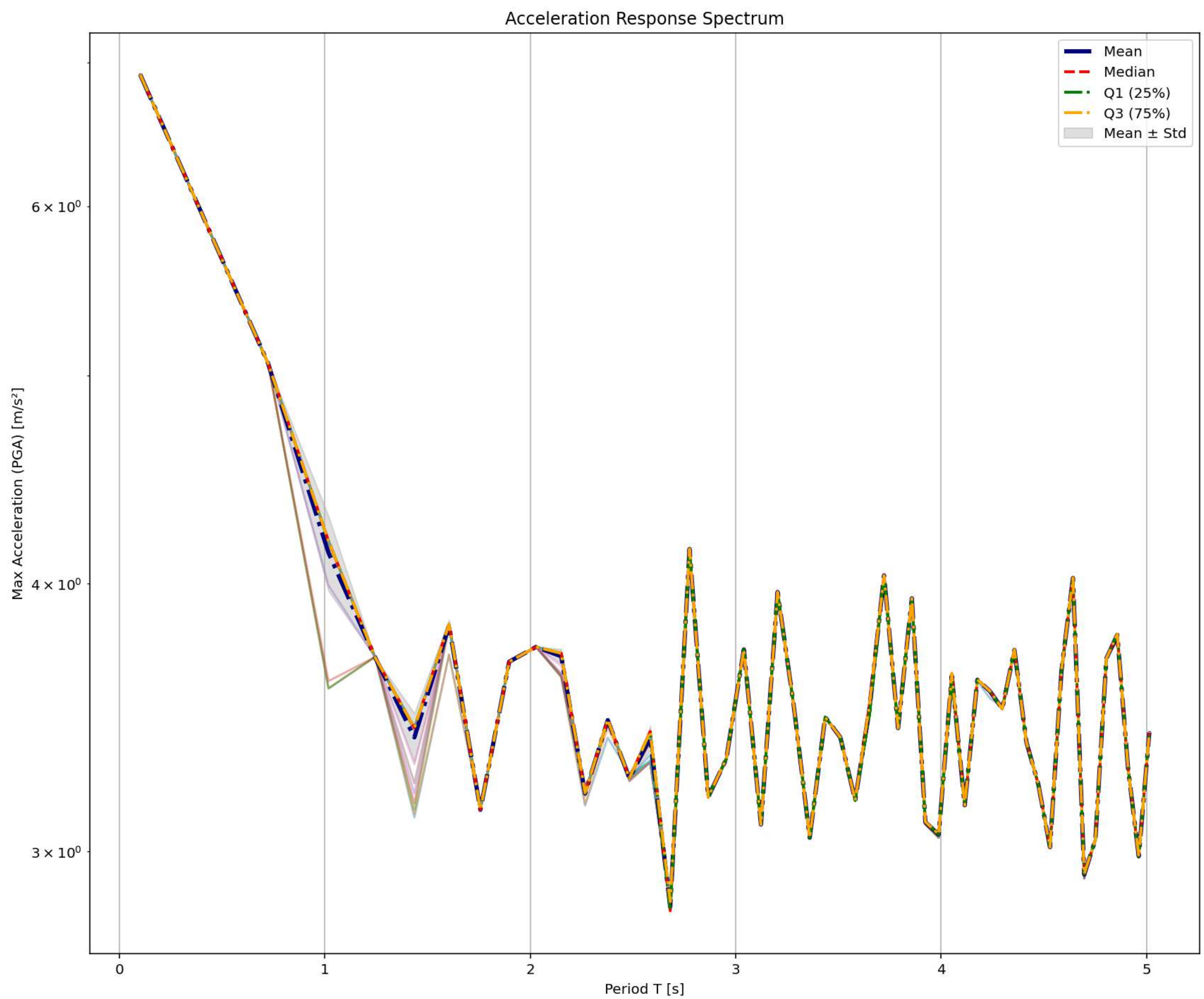

Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
...ESPONSE_SPECTRUM\SDOF_RESPONSE_SPECTRUM_SEISMIC_DUCT_OSF
C:\Users\Dell\Desktop\OPENSEES_FILES\CONSTANT_STR...CT_OSF\SDOF_RESPONSE_SPECTRUM_SEISMIC_DUCT_OSF.py
SDOF_RESPONSE_SPEC...EISMIC_DUCT_OSF.py
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
NONLINEAR DYNAMIC ANALYSIS UNDER MULTI GROUND MOTION RECORDS COMPUTATION AND VISUALIZ
RESPONSE SPECTRA OF ACCELERATION, VELOCITY, DISPLACEMENT DUCTILITY DAMAGE INDEX USING
CONSTANT STRUCTURAL DUCTILITY RATIO RESPONSE SPECTRUM
THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
EMAIL: salar.d.ghashghaei@gmail.com
"""
This code implements a comprehensive nonlinear dynamic analysis framework for
performance-based earthquake engineering assessment of single-degree-of-freedom
(SDOF) systems. The methodology combines traditional nonlinear time-history
analysis with modern probabilistic and machine learning techniques for advanced
structural performance evaluation with changing structural ductility raio and over strength
KEY ENGINEERING OBJECTIVES:
1. Comparative assessment of hysteretic models for seismic response prediction
2. Probabilistic seismic demand analysis using multiple ground motions
3. Development of fragility curves for performance-based earthquake engineering
4. Integration of data science methods for structural reliability assessment
ANALYTICAL FEATURES:
- Nonlinear material behavior with pinching and degradation
- Response spectrum analysis across period range
- Real-time structural health monitoring metrics
- Statistical characterization of seismic demands
- Machine learning-based damage prediction
Model setup:
 - SDOF properties: mass (m), initial stiffness (k), yield displacement (Dy), ultimate disp
 - Hysteresis models: HYSTERETICSM (pinching, stiffness degradation, strength decay).
Displacement Response Spectrum
Mean
Median
Q1 (25%)
Q3 (75%)
Mean ± Std
26 %
IPython Console Files Help Variable Explorer Debugger Plots History
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 554, Col 5 UTF-8 CRLF RW Mem 34%

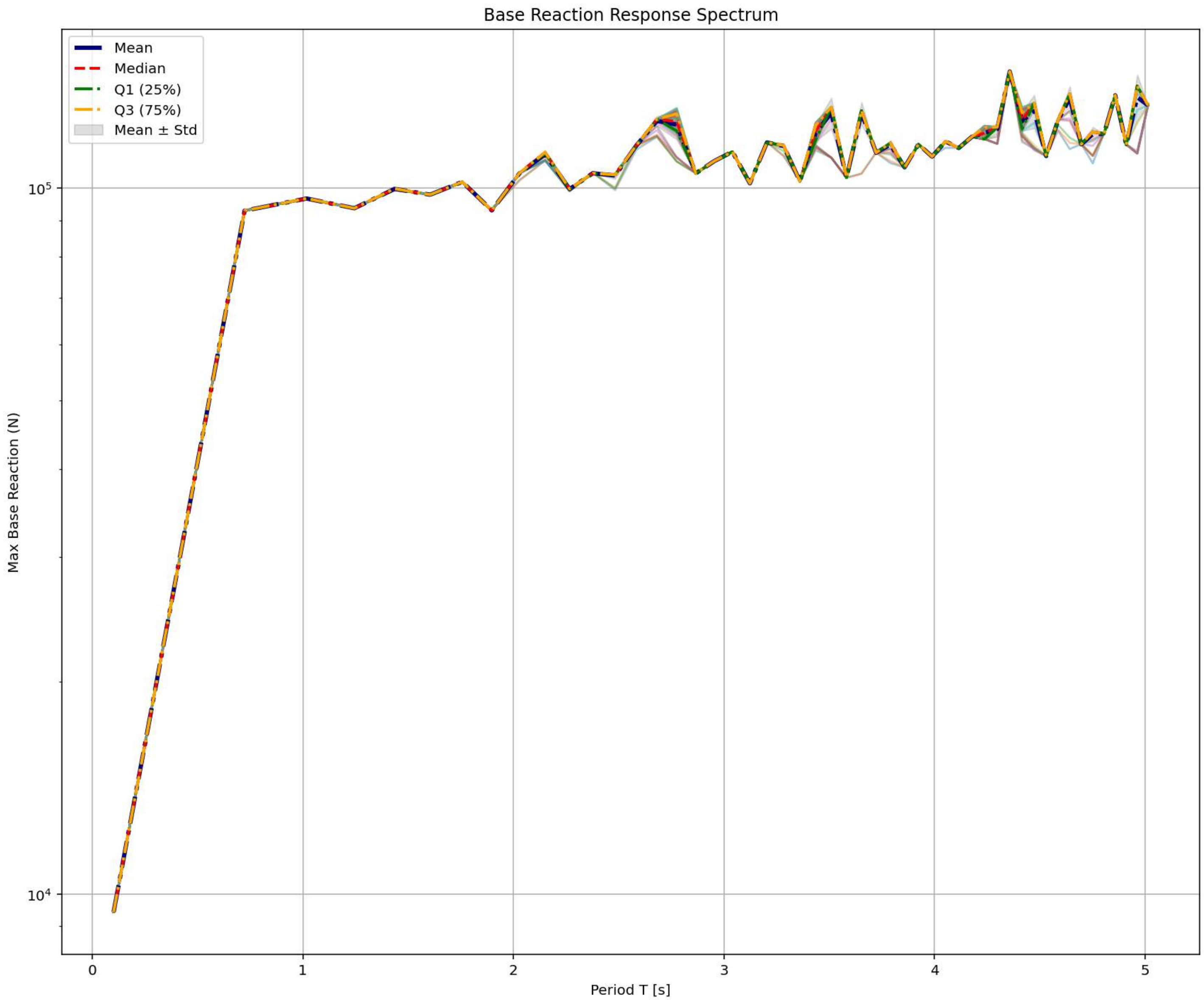


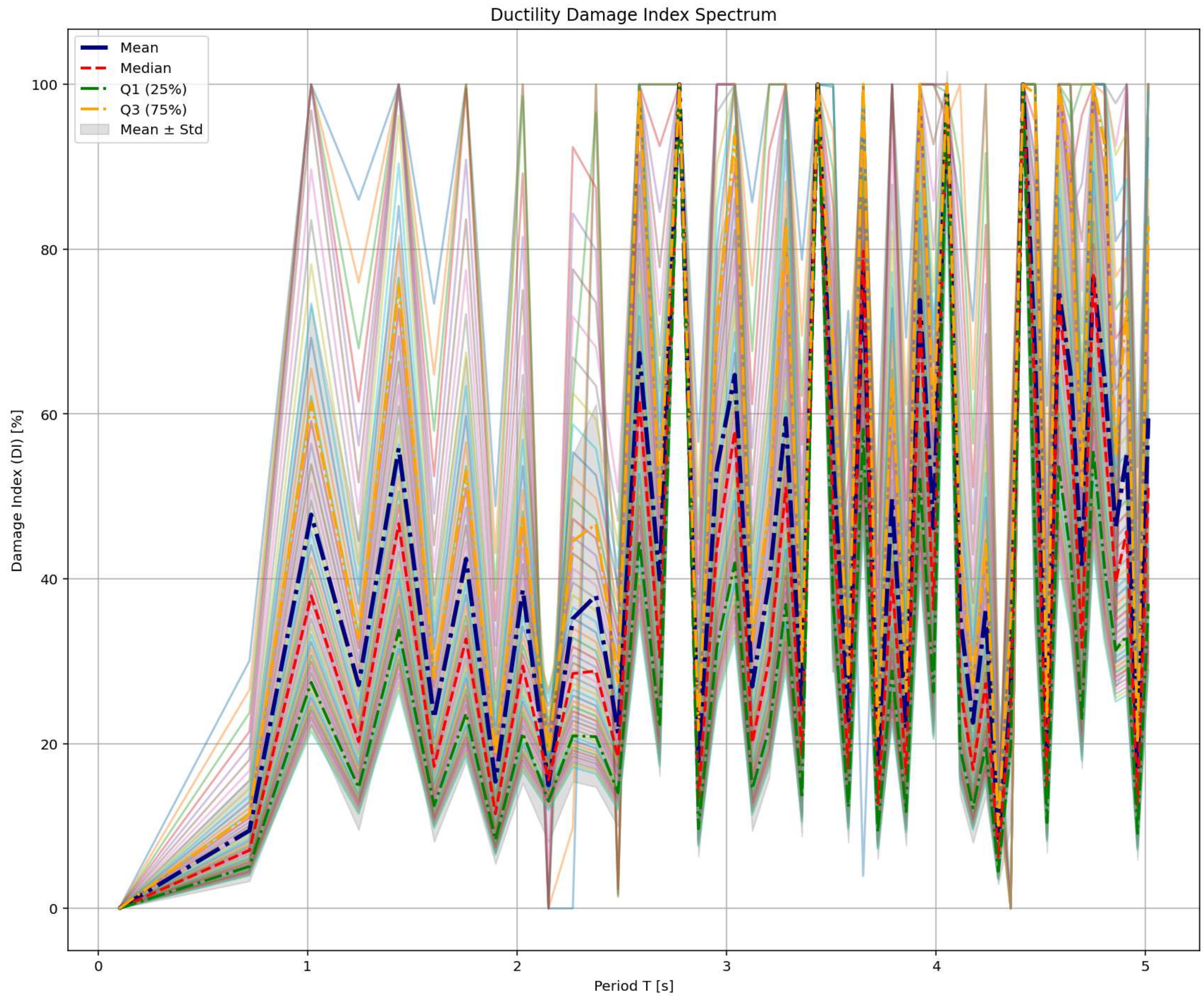
Displacement Response Spectrum
Mean
Median
Q1 (25%)
Q3 (75%)
Mean ± Std
10^0
10^{-1}
10^{-2}
Max Displacement (PGD) [m]
0
1
2
3
4
5
Period T [s]



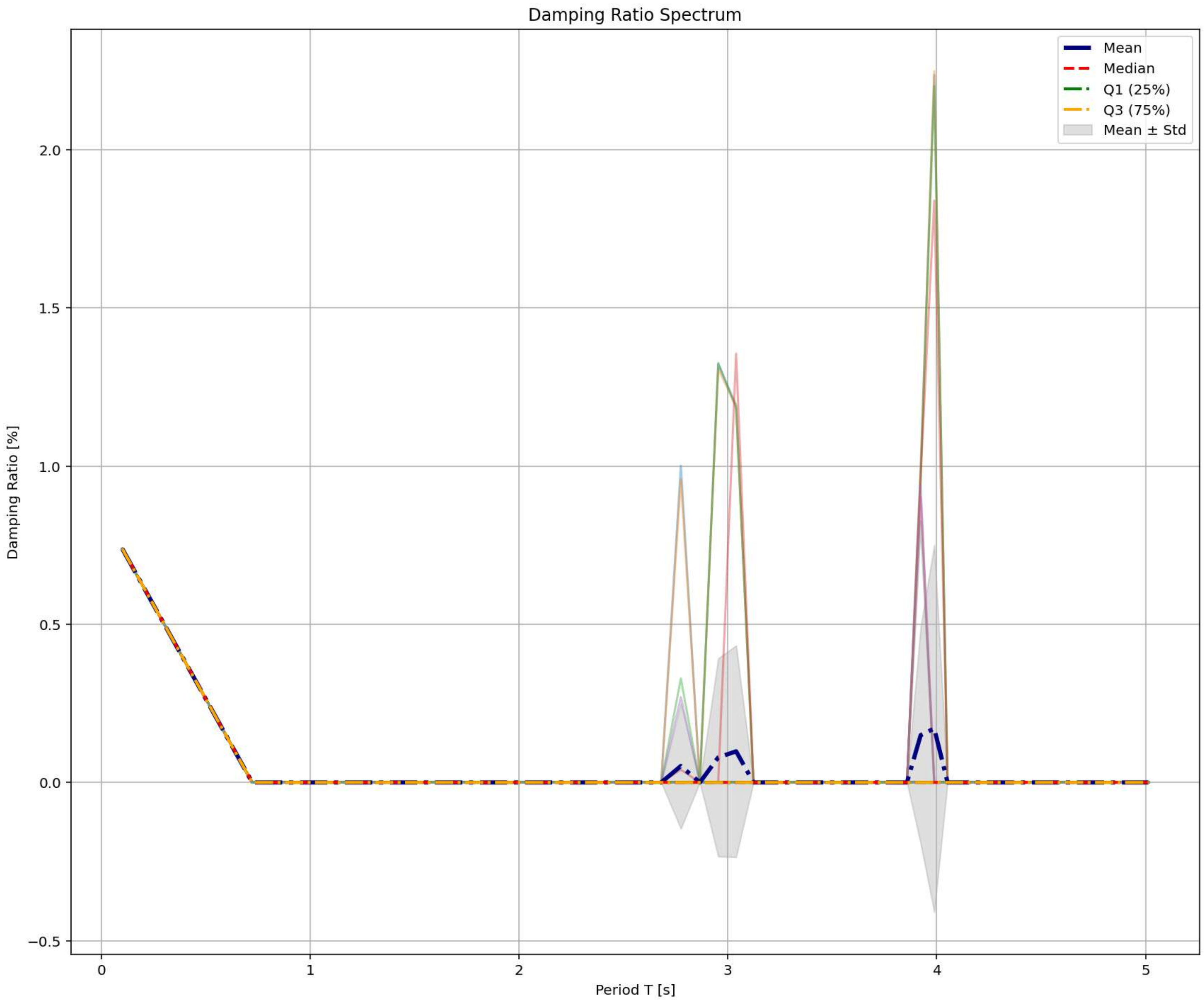
Velocity Response Spectrum
Mean
Median
Q1 (25%)
Q3 (75%)
Mean ± Std
3×10^{-1}
2×10^{-1}
Max Velocity (PGV) [m/s]
0
1
2
3
4
5
Period T [s]

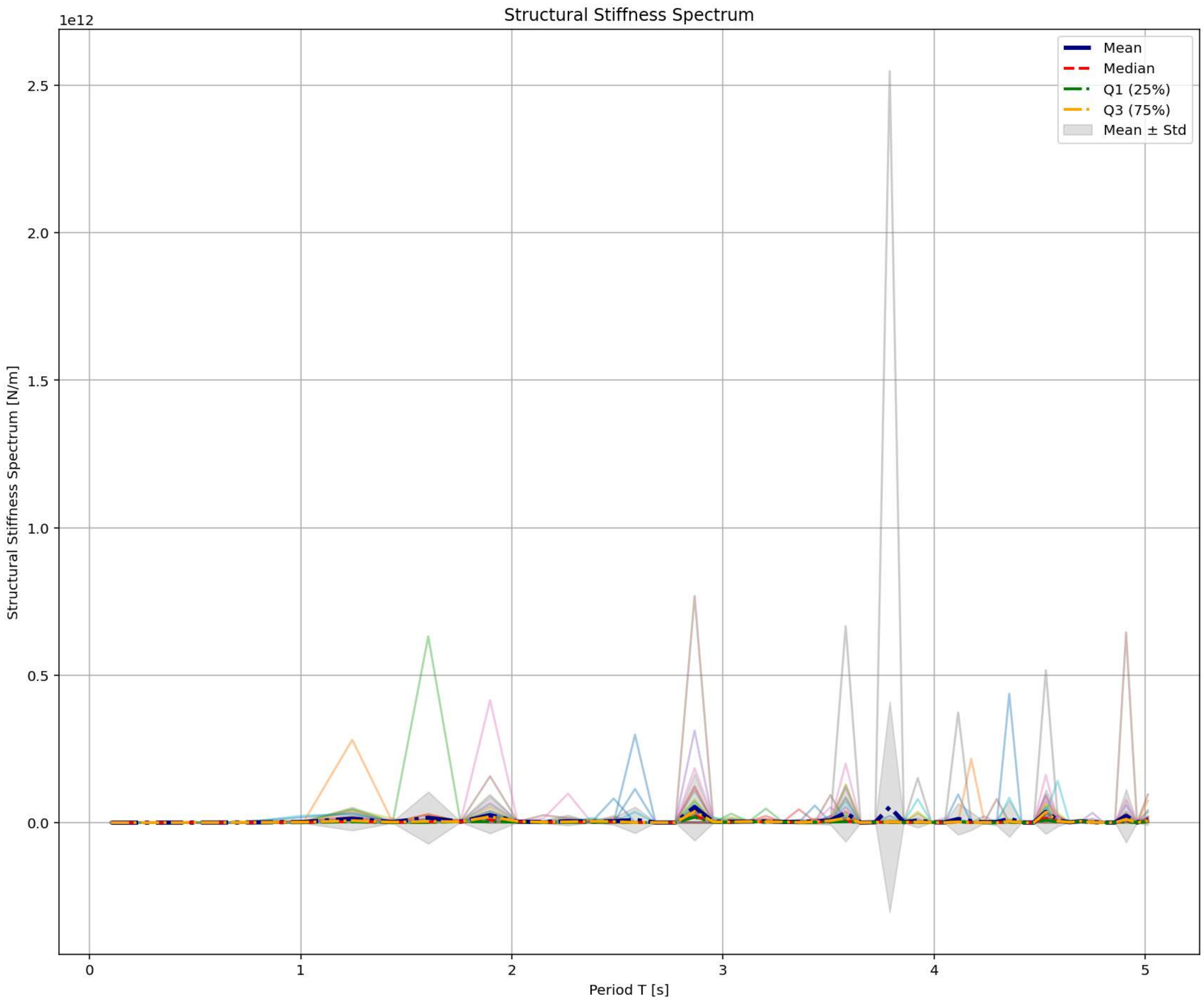

Acceleration Response Spectrum
Mean
Median
Q1 (25%)
Q3 (75%)
Mean ± Std
6×10^0
4×10^0
3×10^0
Max Acceleration (PGA) [m/s²]
0
1
2
3
4
5
Period T [s]

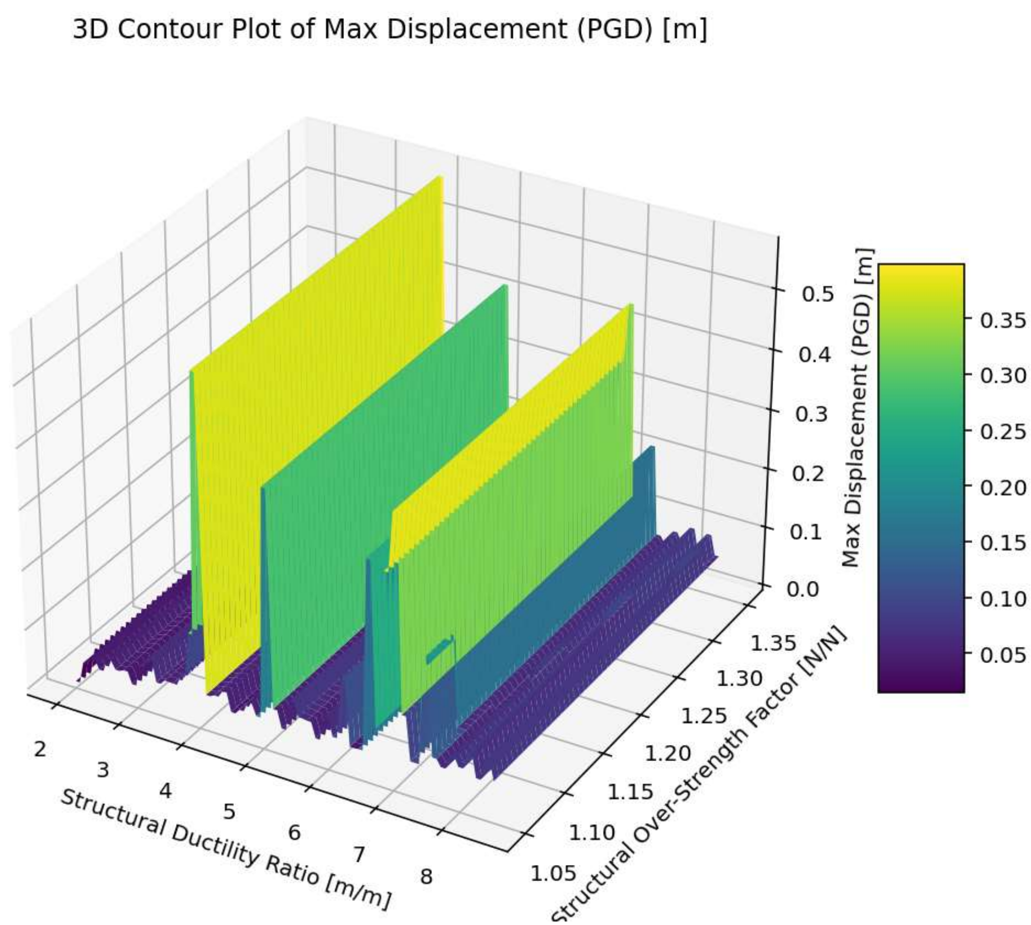

Base Reaction Response Spectrum
Mean
Median
Q1 (25%)
Q3 (75%)
Mean ± Std
Max Base Reaction (N)
10^5
10^4
0
1
2
3
4
5
Period T [s]

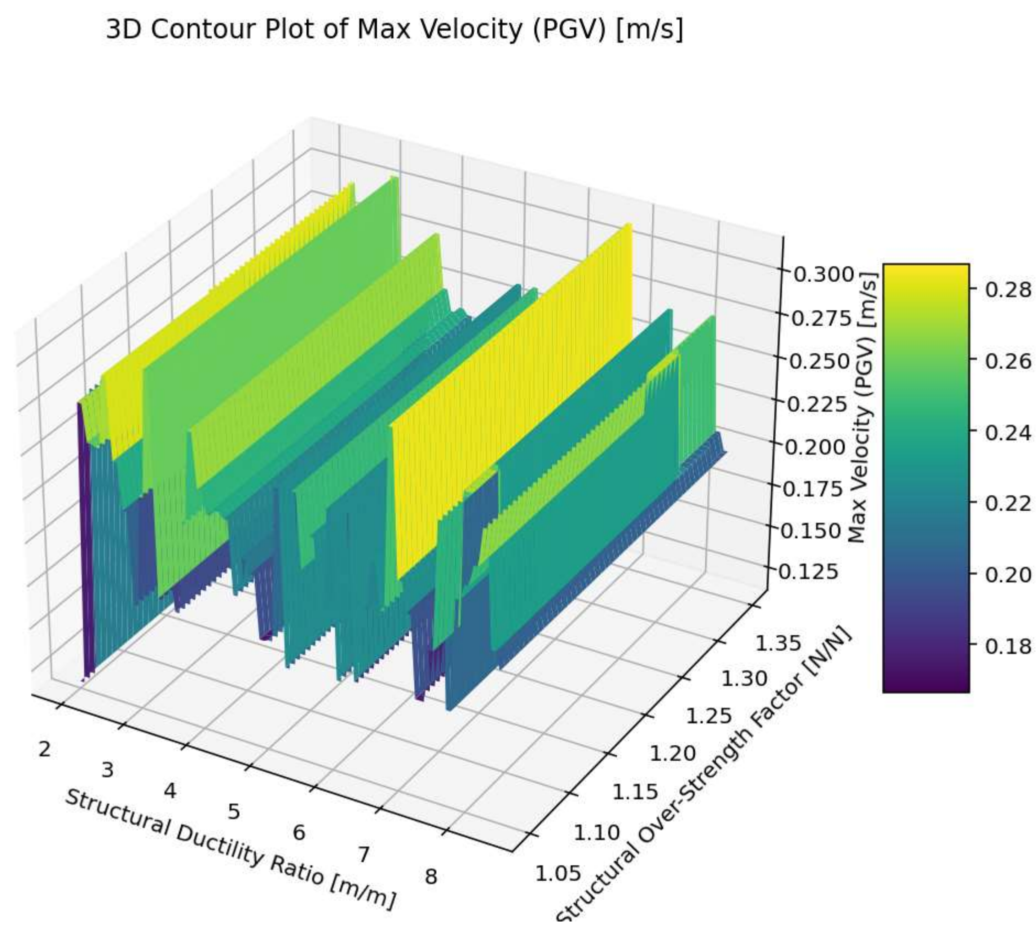


Ductility Damage Index Spectrum
Mean
Median
Q1 (25%)
Q3 (75%)
Mean ± Std
Damage Index (DI) [%]
Period T [s]
0
20
40
60
80
100
0
1
2
3
4
5

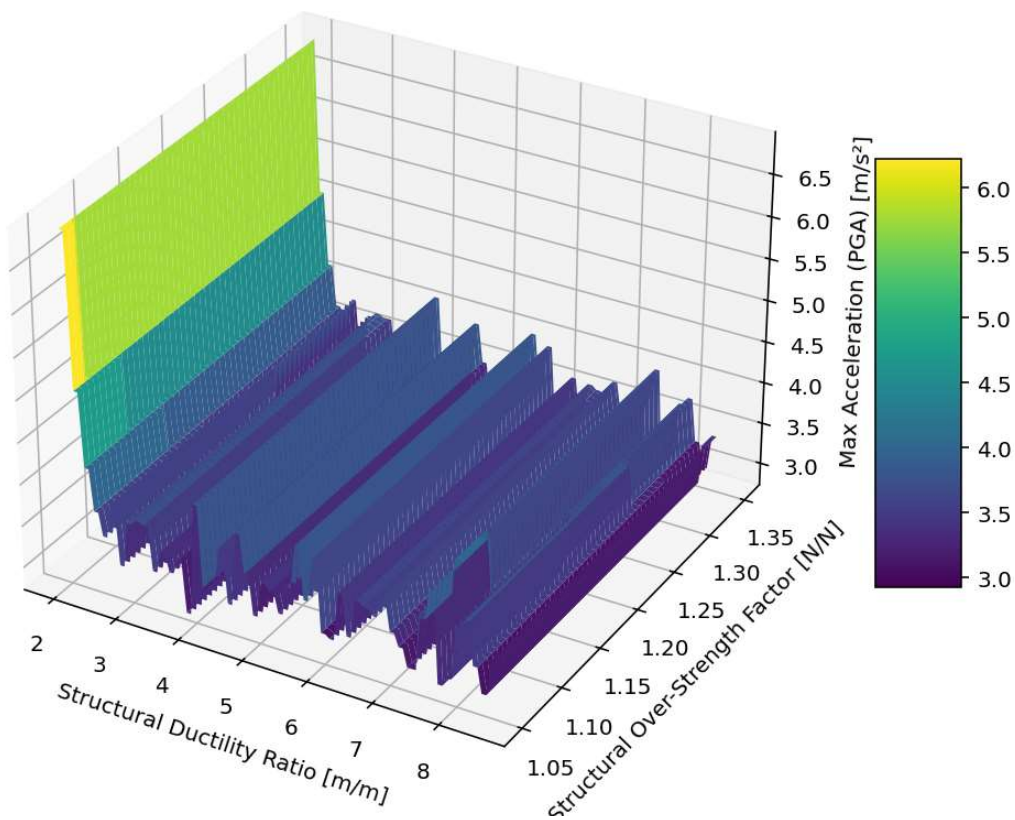

Damping Ratio Spectrum
Mean
Median
Q1 (25%)
Q3 (75%)
Mean ± Std
Damping Ratio [%]
Period T [s]
2.0
1.5
1.0
0.5
0.0
−0.5
0
1
2
3
4
5

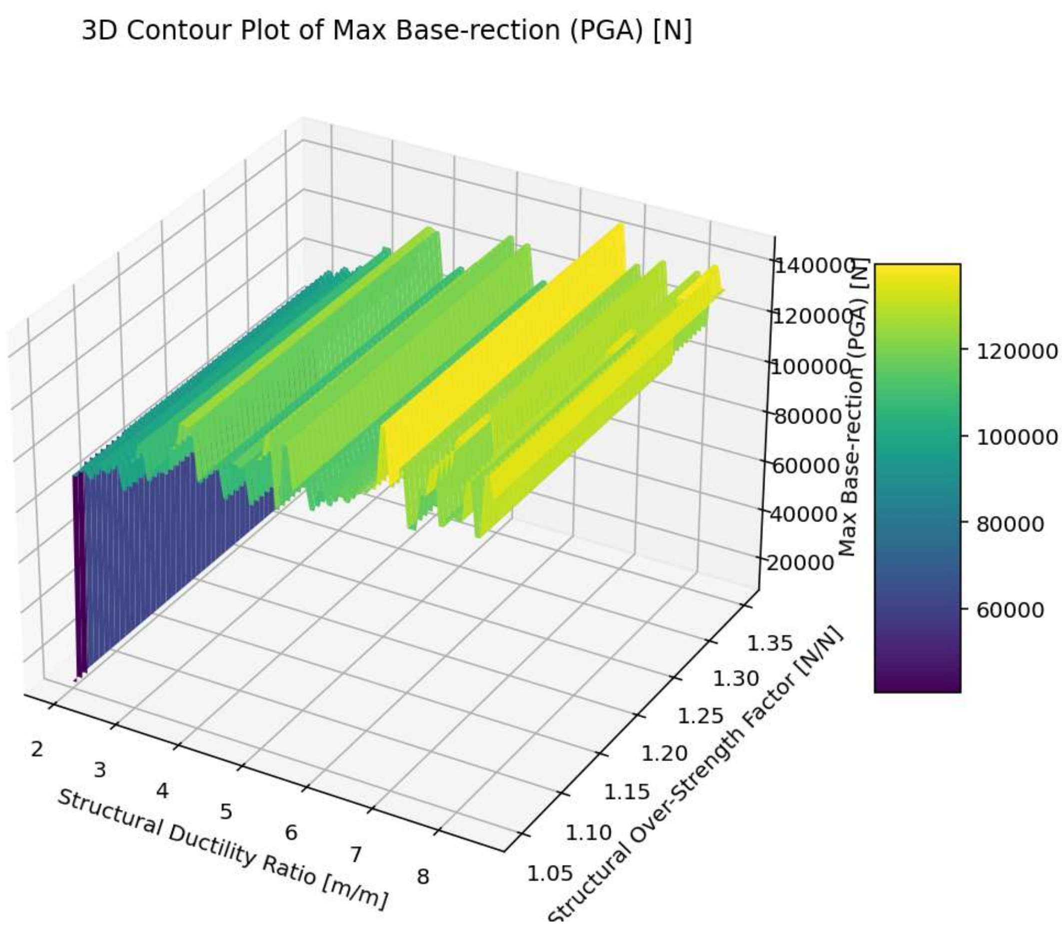

Structural Stiffness Spectrum
1e12
2.5
2.0
1.5
1.0
0.5
0.0
Structural Stiffness Spectrum [N/m]
0
1
2
3
4
5
Period T [s]
Mean
Median
Q1 (25%)
Q3 (75%)
Mean ± Std


3D Contour Plot of Max Displacement (PGD) [m]
Structural Ductility Ratio [m/m]
2
3
4
5
6
7
8
Structural Over-Strength Factor [N/N]
1.05
1.10
1.15
1.20
1.25
1.30
1.35
0.0
0.1
0.2
0.3
0.4
0.5
Max Displacement (PGD) [m]
0.05
0.10
0.15
0.20
0.25
0.30
0.35

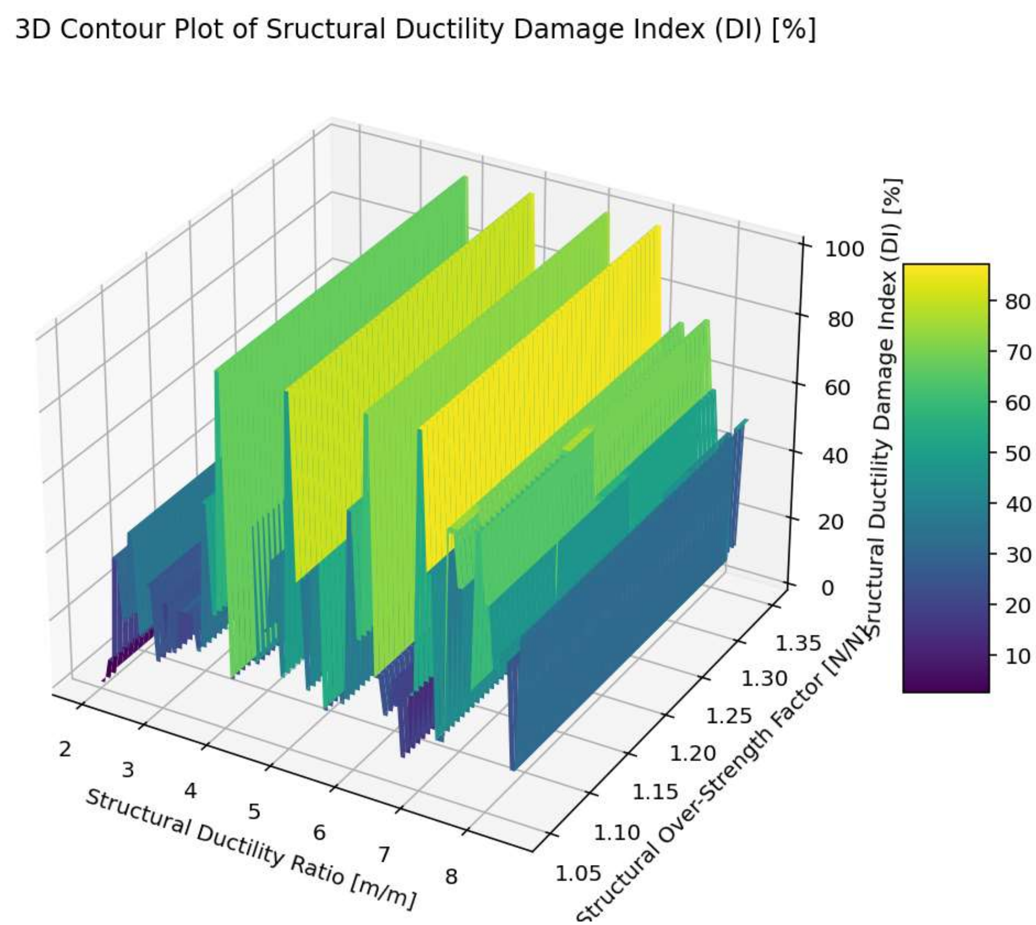

3D Contour Plot of Max Velocity (PGV) [m/s]
Structural Ductility Ratio [m/m]
2
3
4
5
6
7
8
Structural Over-Strength Factor [N/N]
1.05
1.10
1.15
1.20
1.25
1.30
1.35
Max Velocity (PGV) [m/s]
0.125
0.150
0.175
0.200
0.225
0.250
0.275
0.300
0.18
0.20
0.22
0.24
0.26
0.28

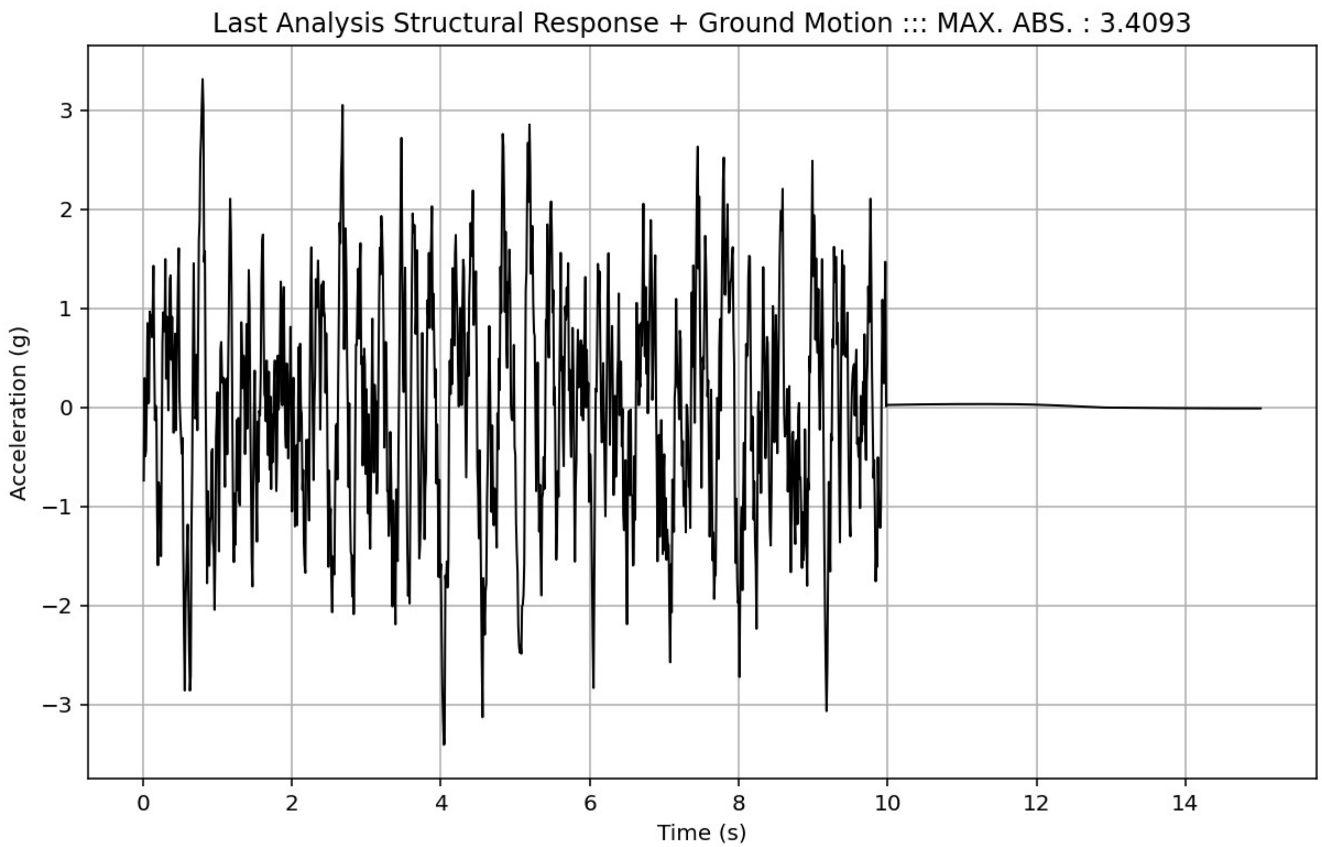
3D Contour Plot of Max Acceleration (PGA) [m/s²]

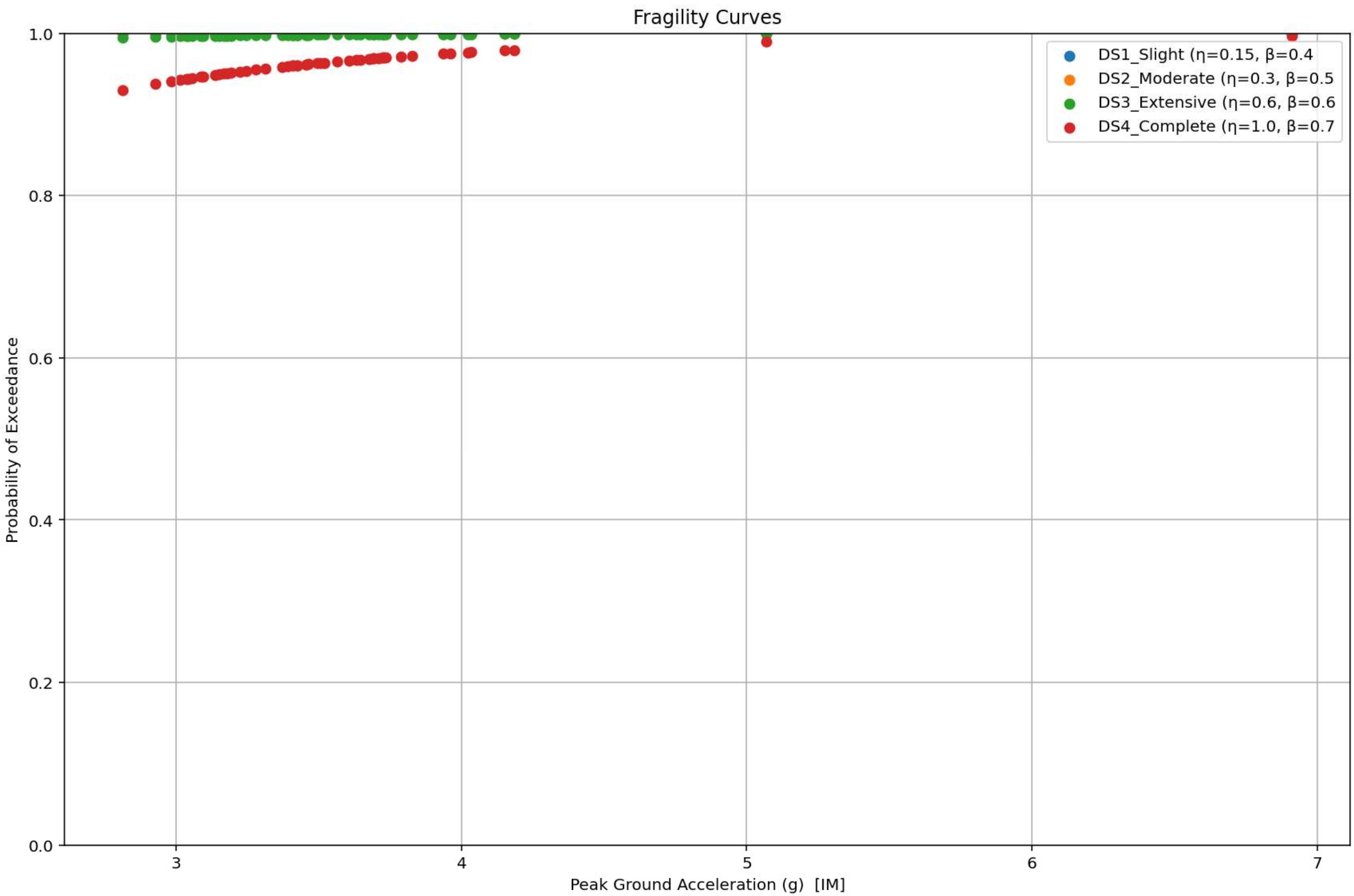


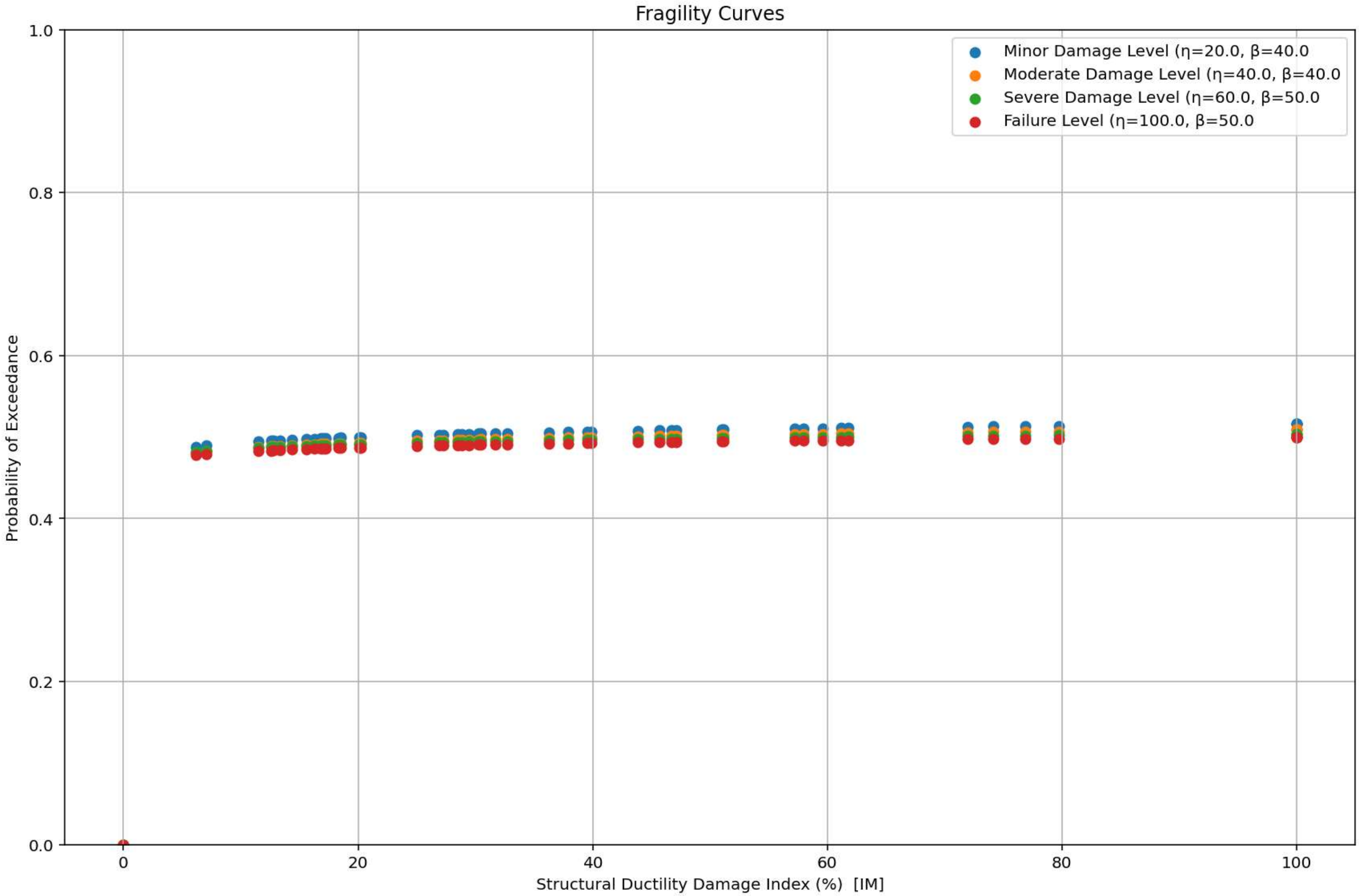


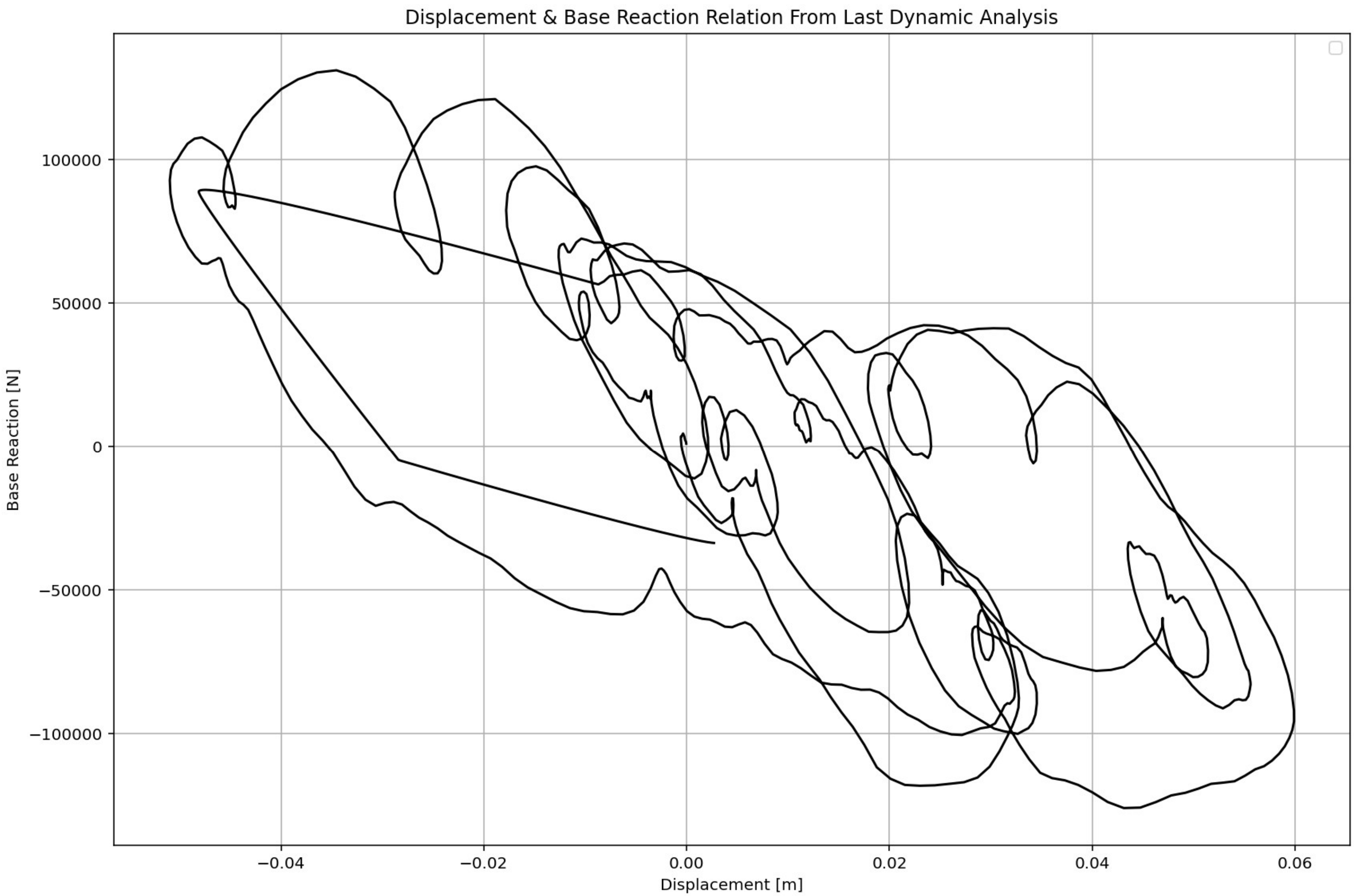
3D Contour Plot of Max Base-rection (PGA) [N]
Structural Ductility Ratio [m/m]
2
3
4
5
6
7
8
Structural Over-Strength Factor [N/N]
1.05
1.10
1.15
1.20
1.25
1.30
1.35
Max Base-rection (PGA) [N]
20000
40000
60000
80000
100000
120000
140000
60000
80000
100000
120000


3D Contour Plot of Sructural Ductility Damage Index (DI) [%]
Structural Ductility Ratio [m/m]
2
3
4
5
6
7
8
Structural Over-Strength Factor [N/N]
1.05
1.10
1.15
1.20
1.25
1.30
1.35
Sructural Ductility Damage Index (DI) [%]
0
20
40
60
80
100
10
20
30
40
50
60
70
80

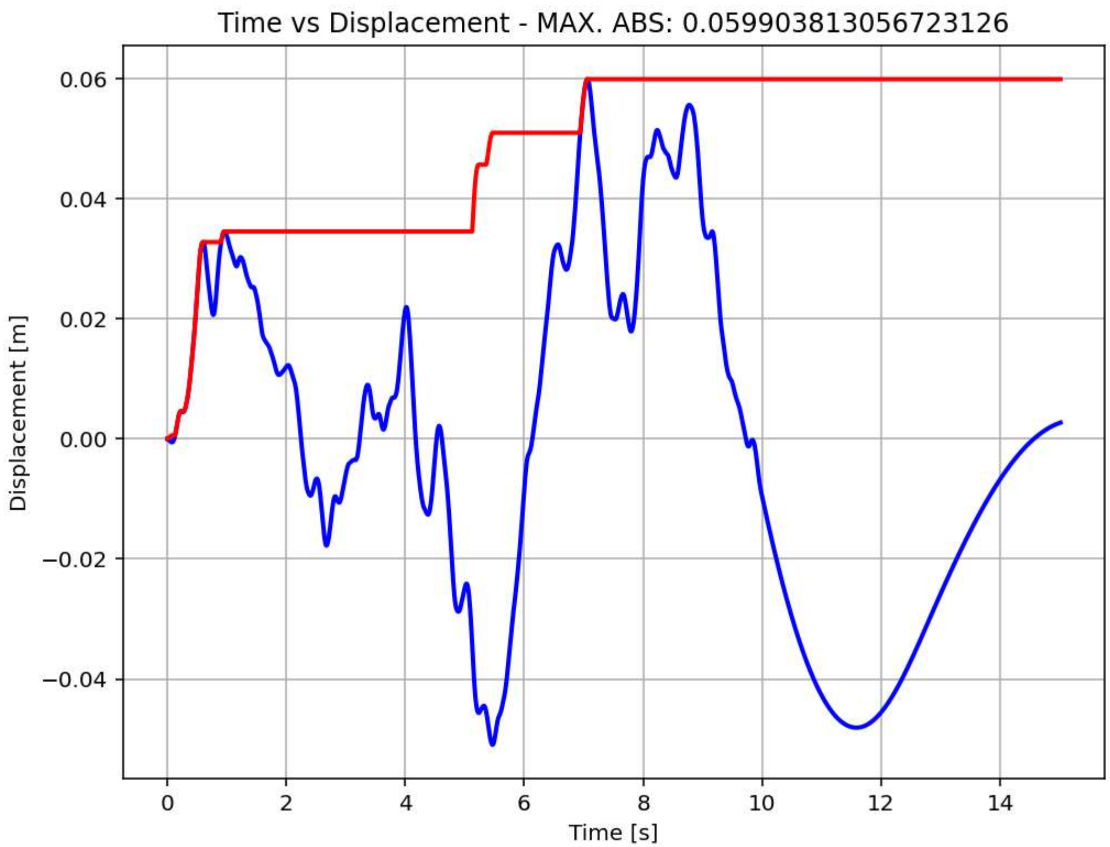

Last Analysis Structural Response + Ground Motion ::: MAX. ABS. : 3.4093
Acceleration (g)
Time (s)
3
2
1
0
−1
−2
−3
0
2
4
6
8
10
12
14


Fragility Curves
DS1_Slight (η=0.15, β=0.4
DS2_Moderate (η=0.3, β=0.5
DS3_Extensive (η=0.6, β=0.6
DS4_Complete (η=1.0, β=0.7
Probability of Exceedance
Peak Ground Acceleration (g) [IM]
0.0
0.2
0.4
0.6
0.8
1.0
3
4
5
6
7

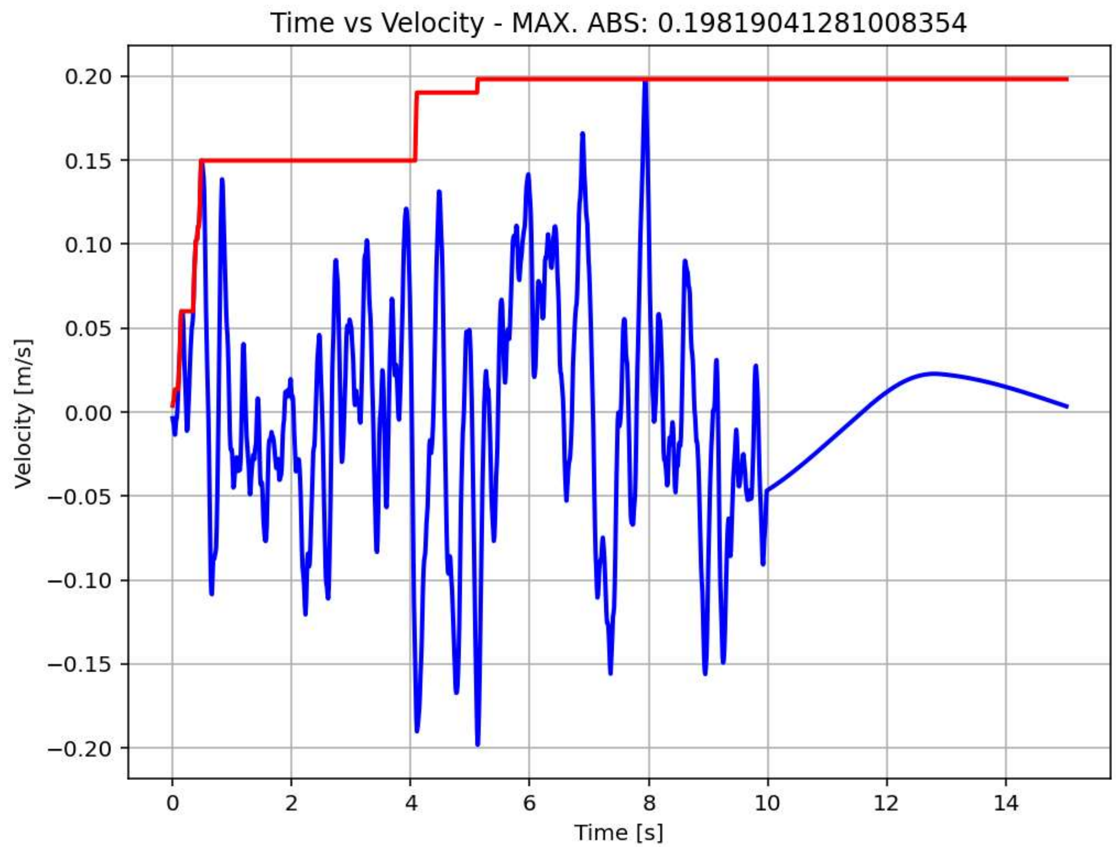

Fragility Curves
Minor Damage Level (η=20.0, β=40.0
Moderate Damage Level (η=40.0, β=40.0
Severe Damage Level (η=60.0, β=50.0
Failure Level (η=100.0, β=50.0
Probability of Exceedance
Structural Ductility Damage Index (%) [IM]
0.0
0.2
0.4
0.6
0.8
1.0
0
20
40
60
80
100

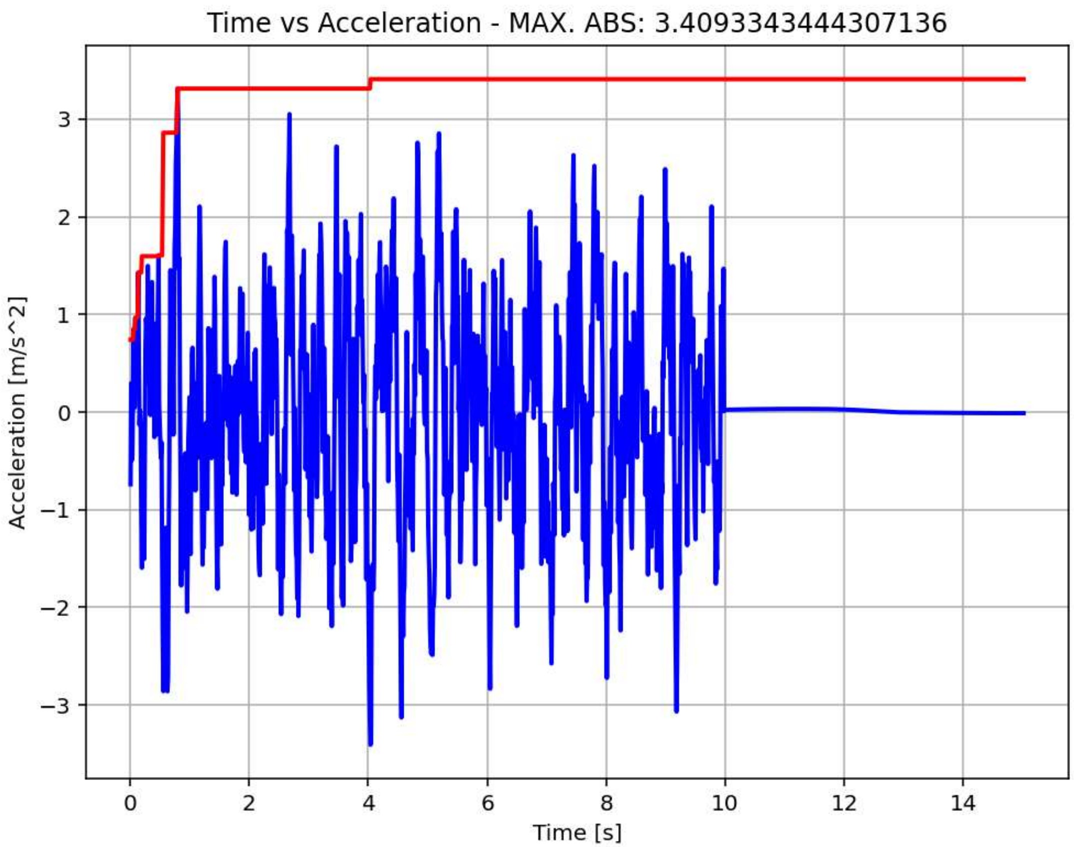


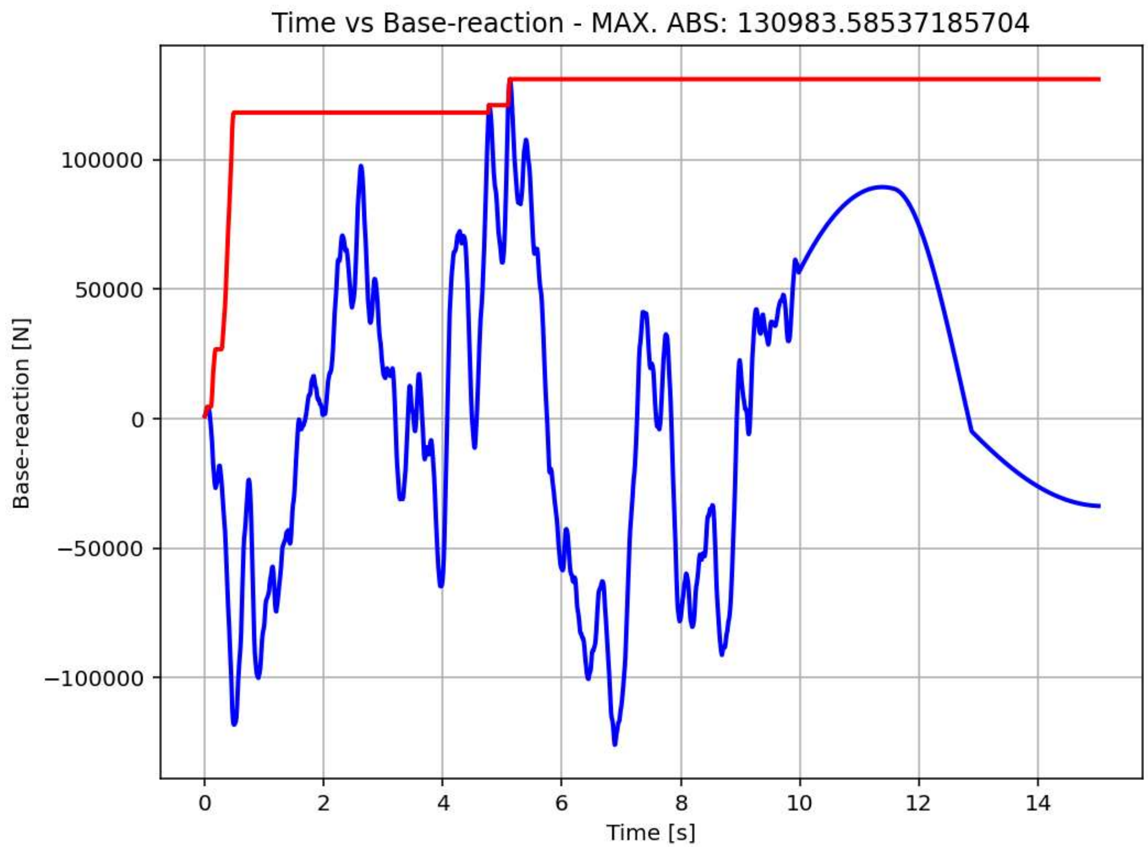
Displacement & Base Reaction Relation From Last Dynamic Analysis
100000
50000
0
−50000
−100000
Base Reaction [N]
−0.04
−0.02
0.00
0.02
0.04
0.06
Displacement [m]

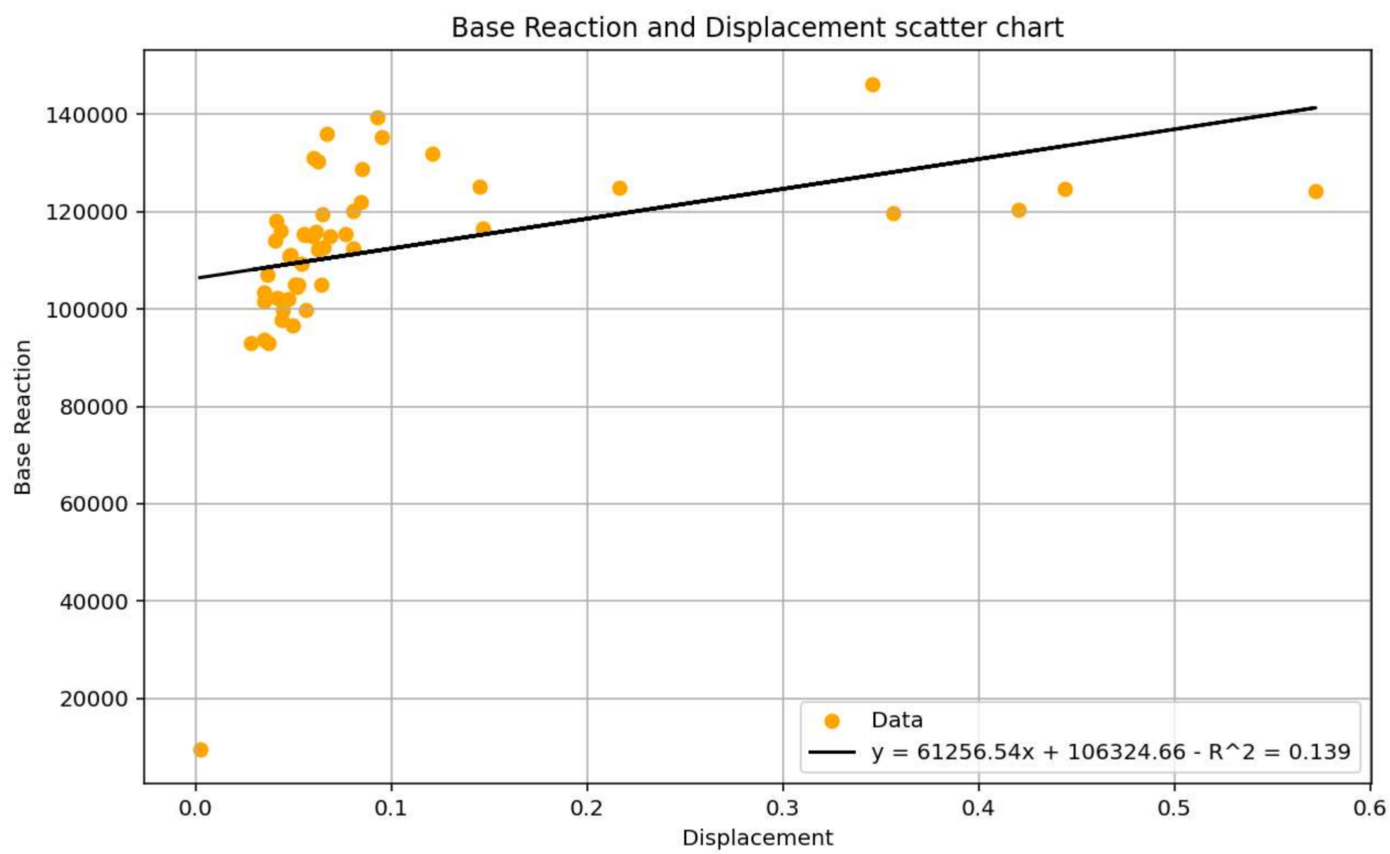


Time vs Displacement - MAX. ABS: 0.059903813056723126
Displacement [m]
Time [s]
0.06
0.04
0.02
0.00
−0.02
−0.04
0
2
4
6
8
10
12
14

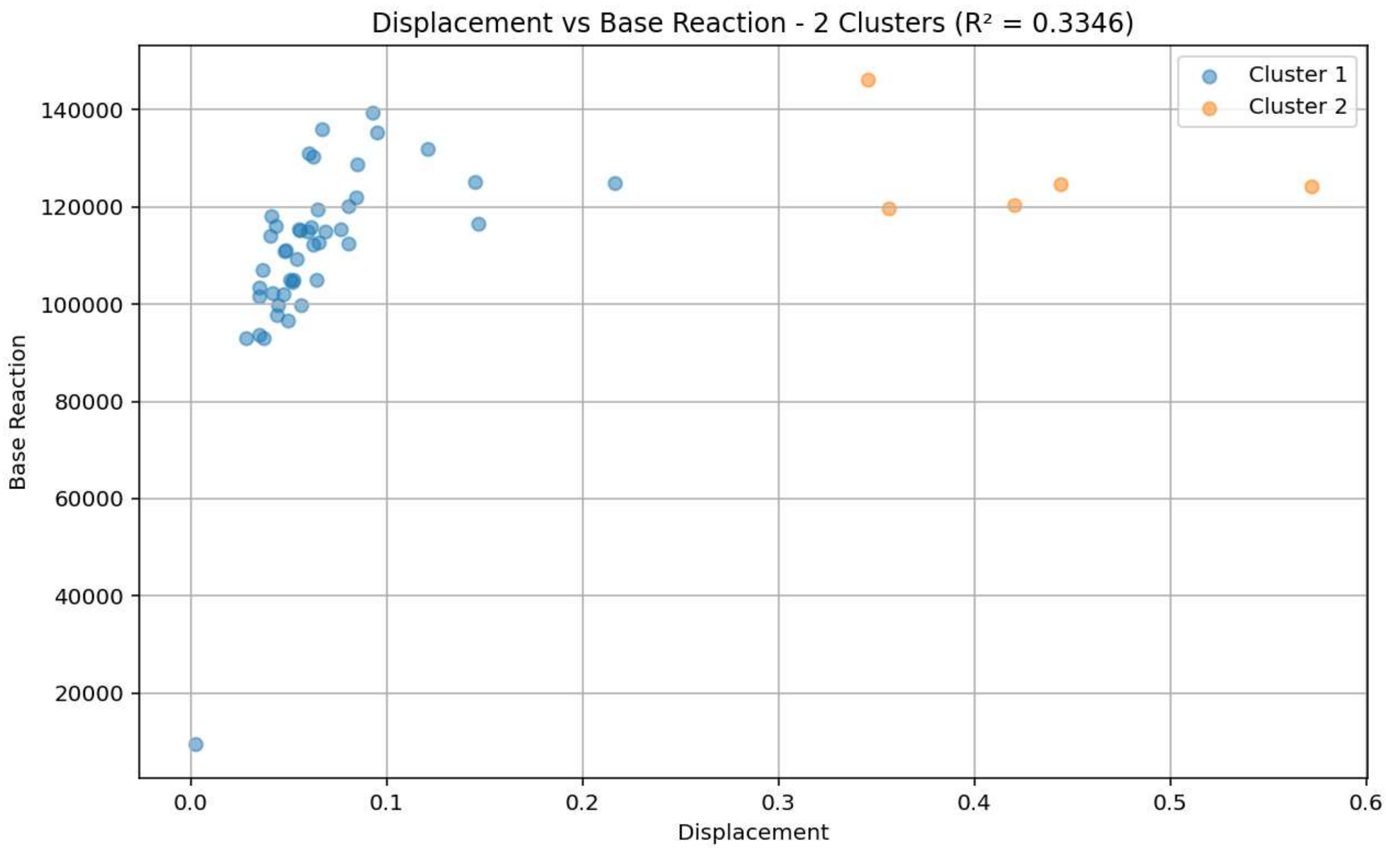

Time vs Velocity - MAX. ABS: 0.19819041281008354
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
Velocity [m/s]
0
2
4
6
8
10
12
14
Time [s]

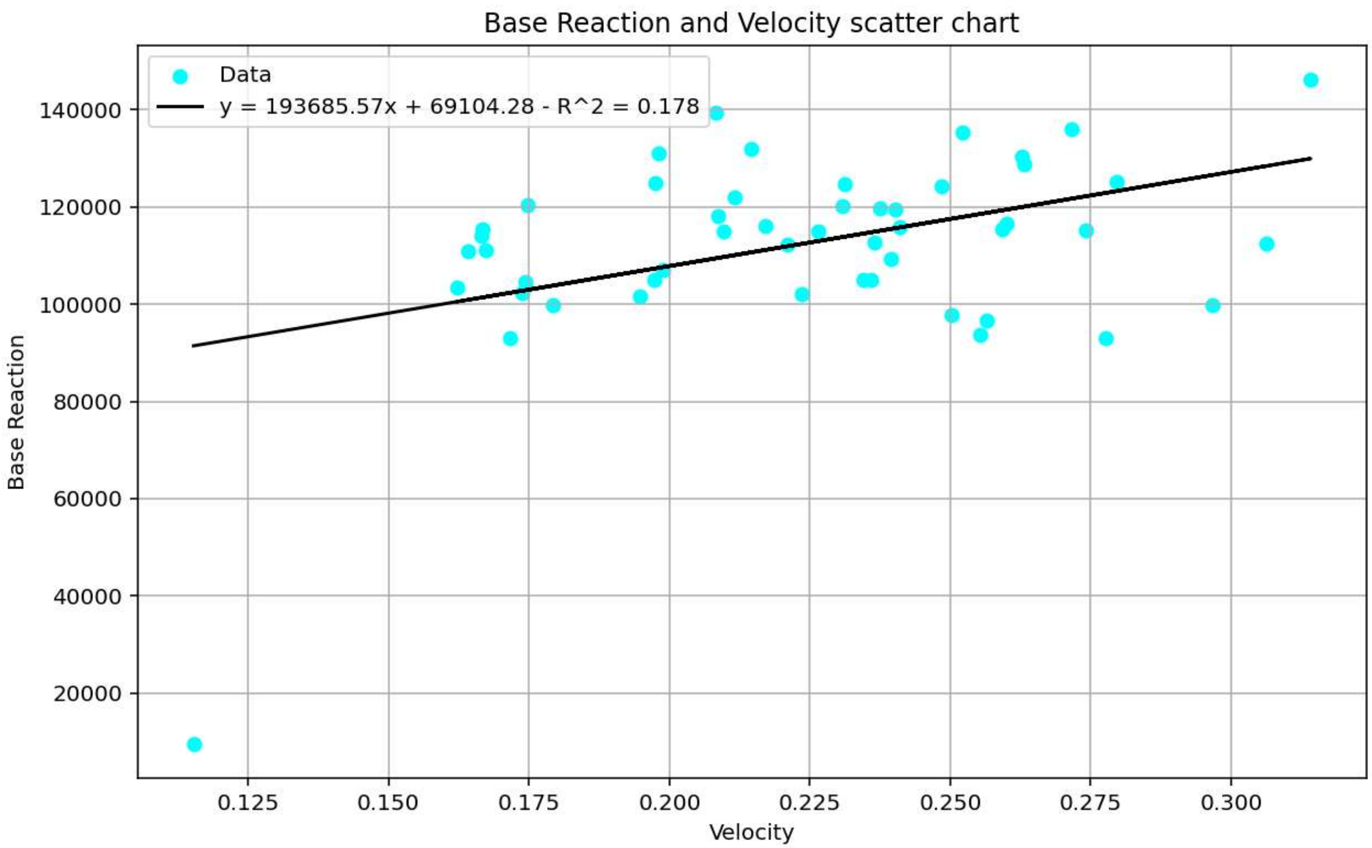

Time vs Acceleration - MAX. ABS: 3.4093343444307136
Acceleration [m/s^2]
Time [s]
3
2
1
0
−1
−2
−3
0
2
4
6
8
10
12
14

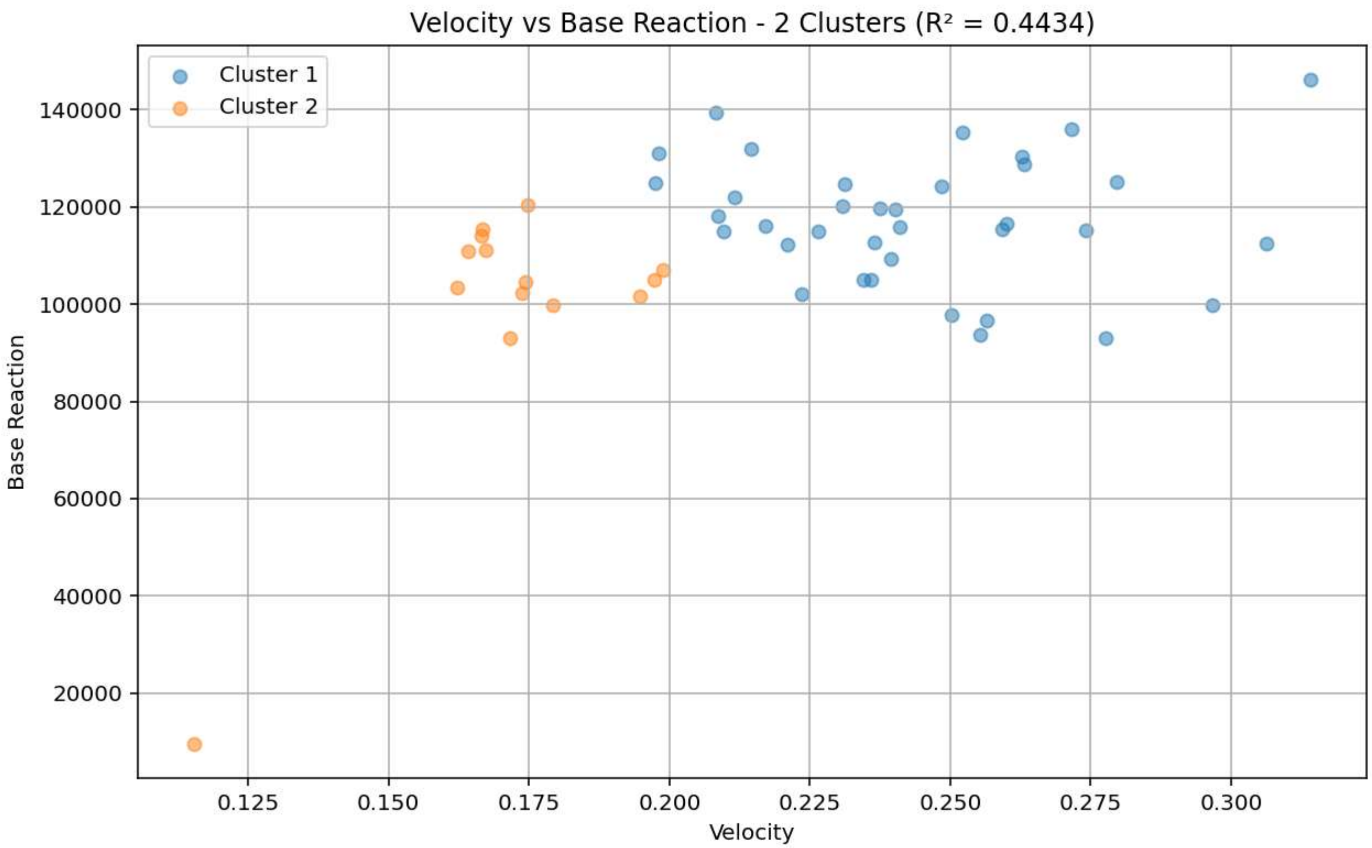

Time vs Base-reaction - MAX. ABS: 130983.58537185704
Base-reaction [N]
100000
50000
0
−50000
−100000
Time [s]
0
2
4
6
8
10
12
14

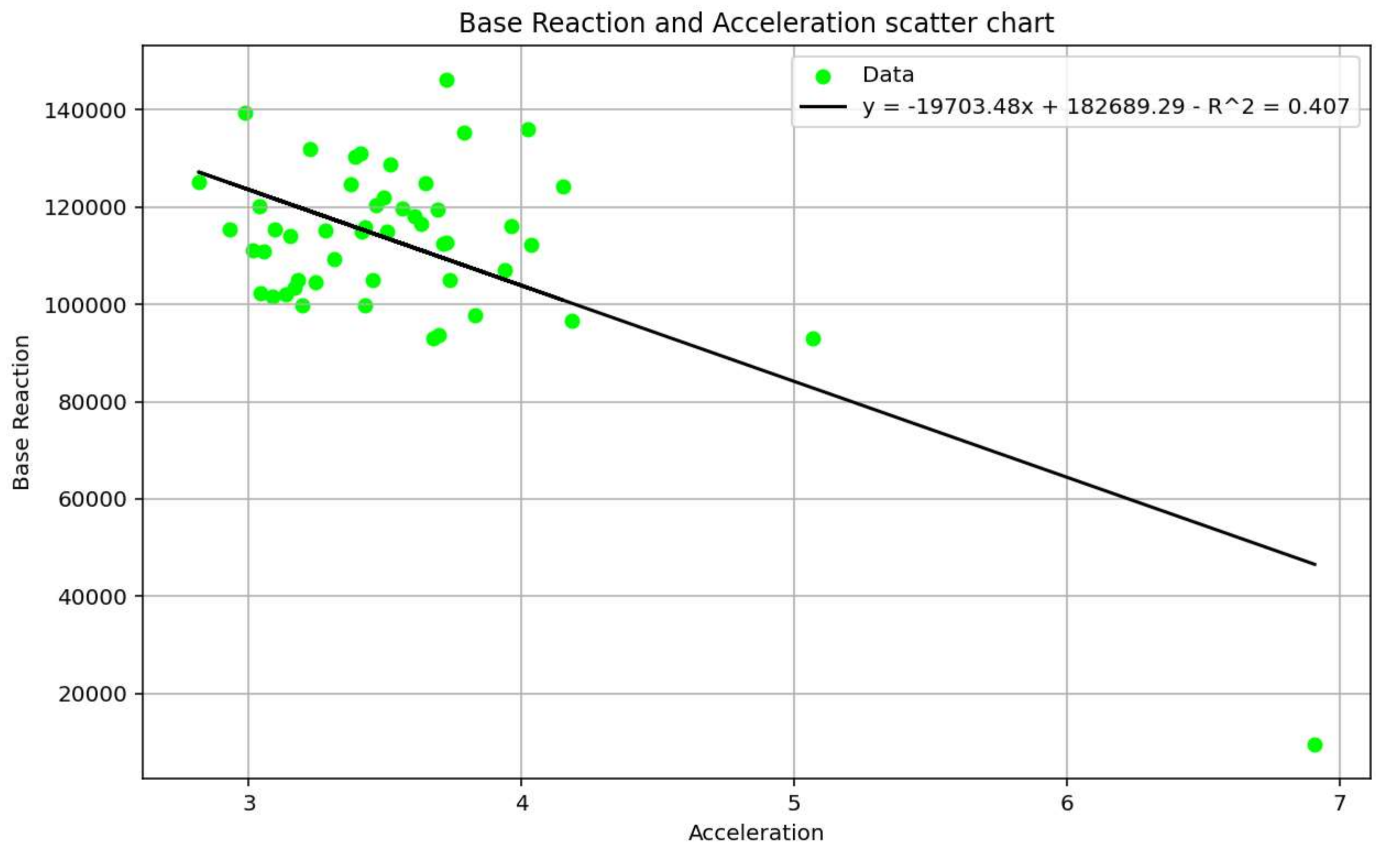


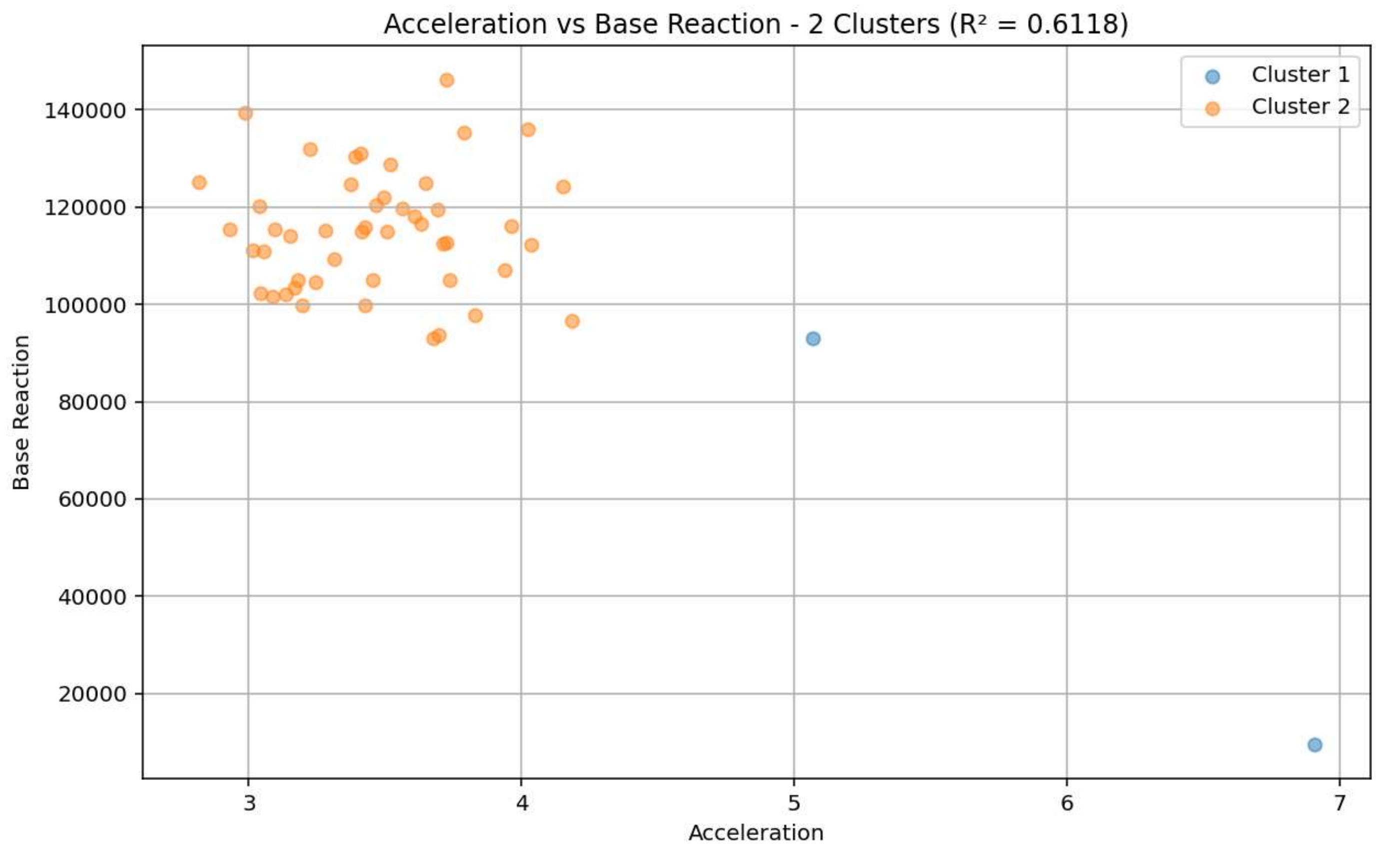
Base Reaction and Displacement scatter chart
140000
120000
100000
80000
60000
40000
20000
Base Reaction
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Displacement
Data
y = 61256.54x + 106324.66 - R^2 = 0.139


Displacement vs Base Reaction - 2 Clusters (R² = 0.3346)
Cluster 1
Cluster 2
Base Reaction
Displacement
140000
120000
100000
80000
60000
40000
20000
0.0
0.1
0.2
0.3
0.4
0.5
0.6

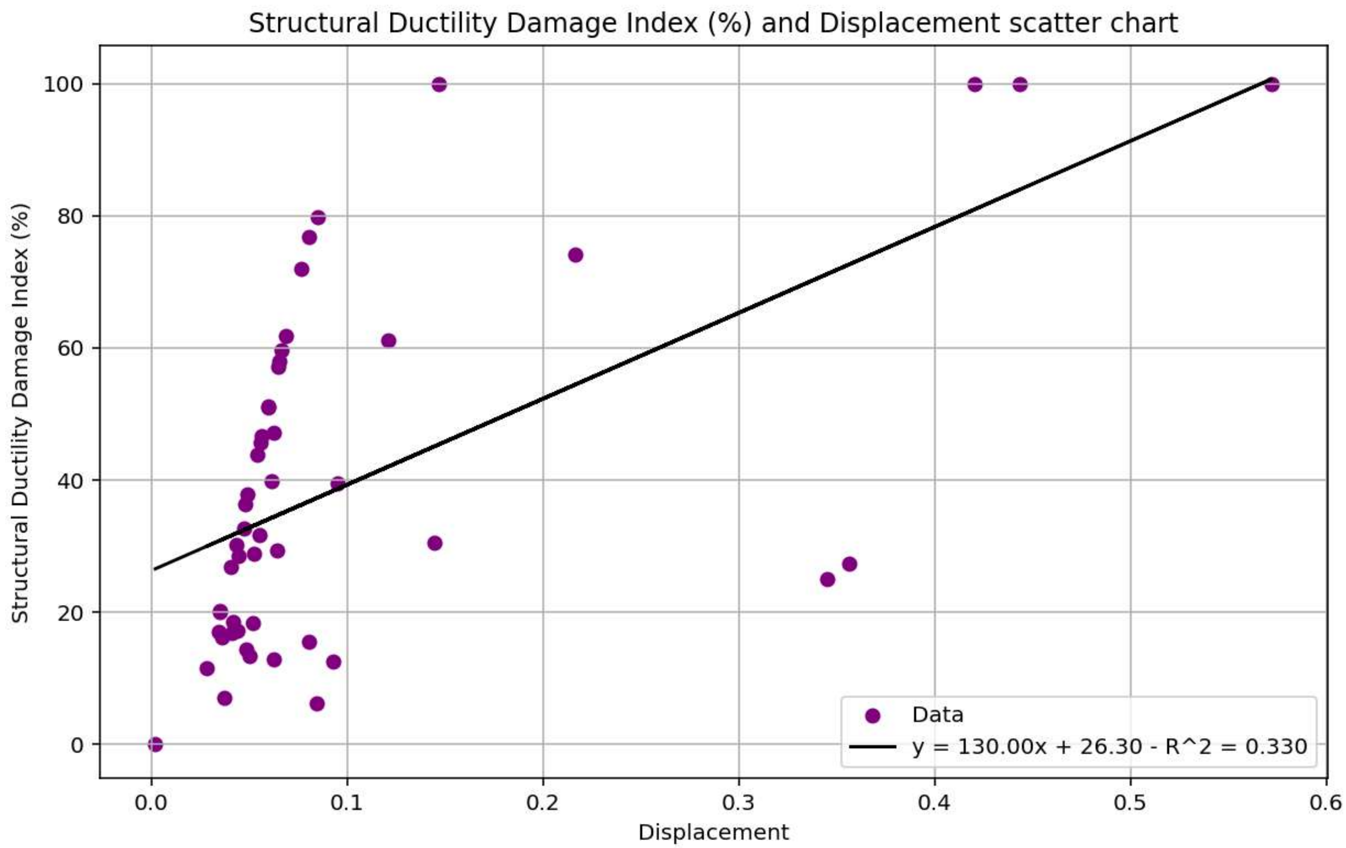

Base Reaction and Velocity scatter chart
Data
y = 193685.57x + 69104.28 - R^2 = 0.178
Base Reaction
140000
120000
100000
80000
60000
40000
20000
0.125
0.150
0.175
0.200
0.225
0.250
0.275
0.300
Velocity

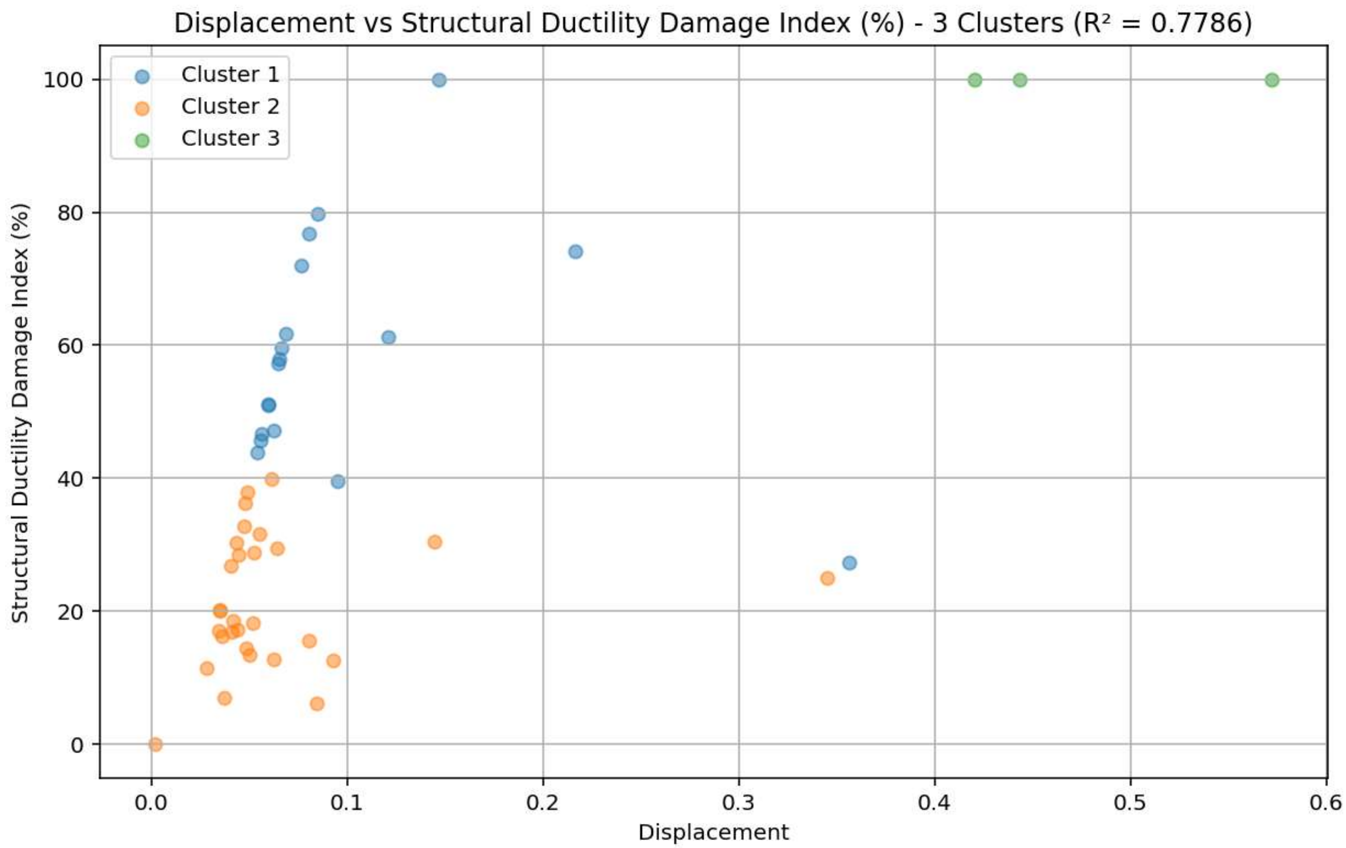

Velocity vs Base Reaction - 2 Clusters (R^2 = 0.4434)
Cluster 1
Cluster 2
140000
120000
100000
80000
60000
40000
20000
Base Reaction
0.125
0.150
0.175
0.200
0.225
0.250
0.275
0.300
Velocity


Base Reaction and Acceleration scatter chart
Data
y = -19703.48x + 182689.29 - R^2 = 0.407
Base Reaction
140000
120000
100000
80000
60000
40000
20000
3
4
5
6
7
Acceleration

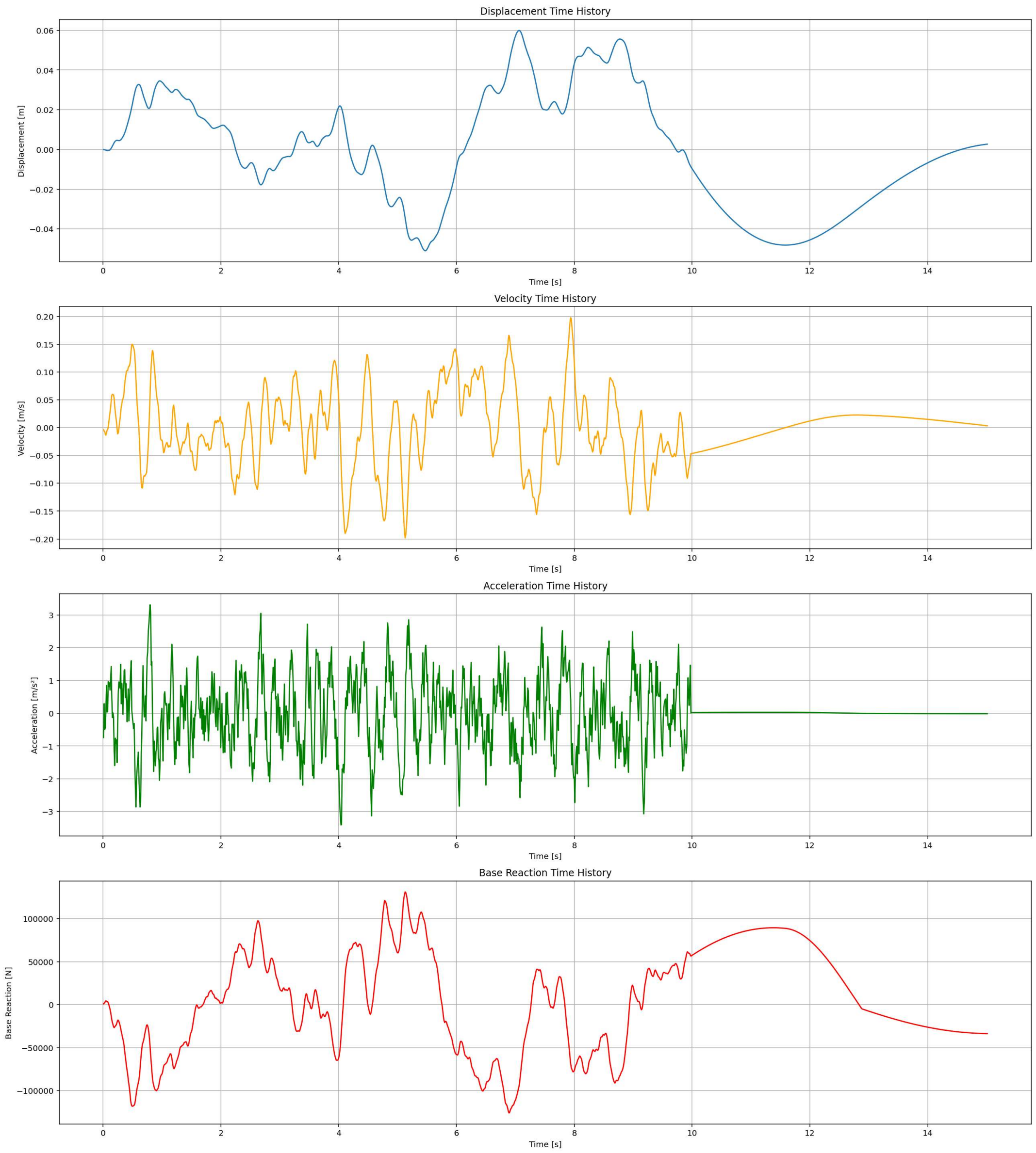


Acceleration vs Base Reaction - 2 Clusters (R² = 0.6118)
Cluster 1
Cluster 2
140000
120000
100000
80000
60000
40000
20000
Base Reaction
3
4
5
6
7
Acceleration

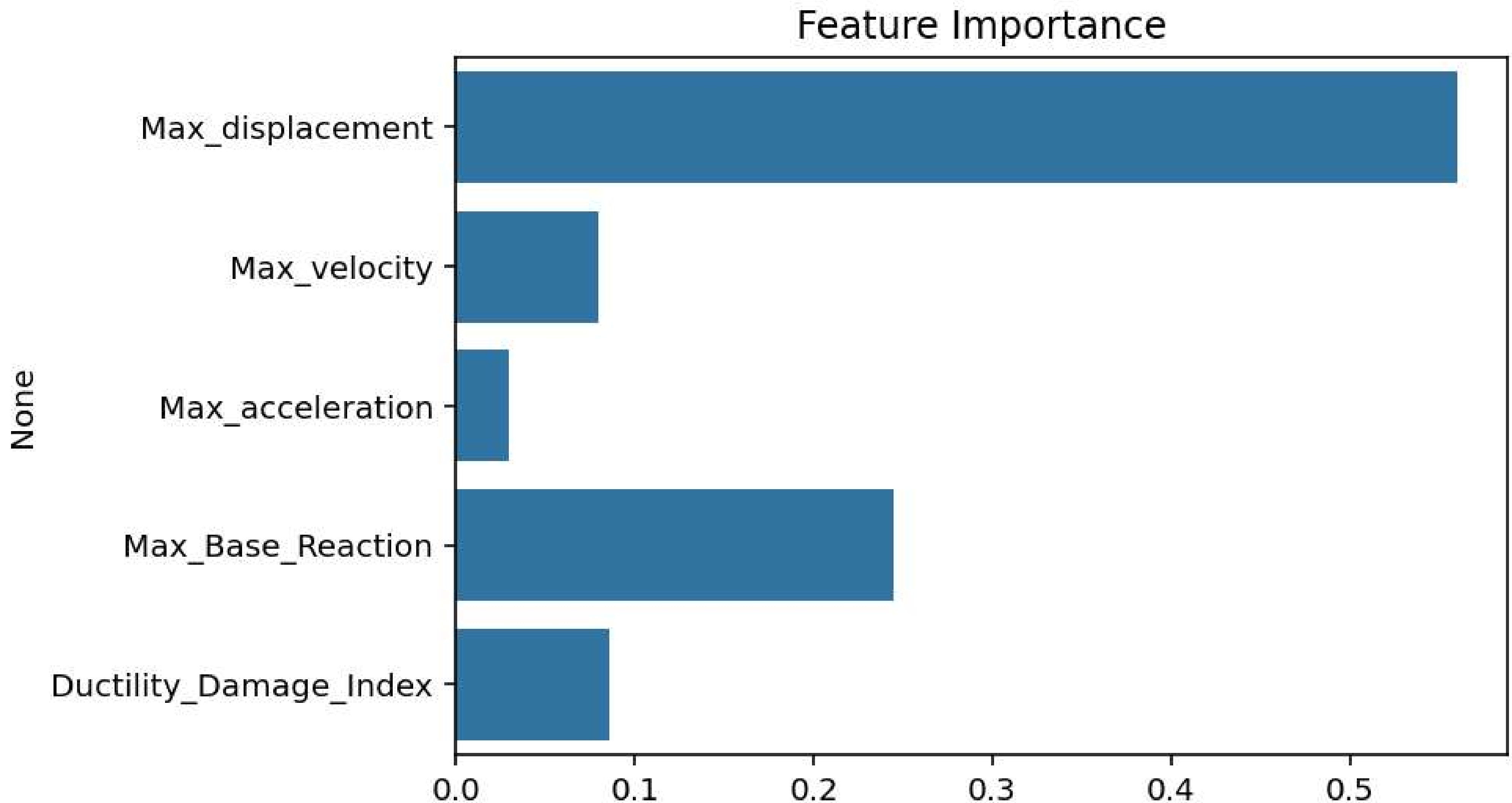

Structural Ductility Damage Index (%) and Displacement scatter chart
Structural Ductility Damage Index (%)
Displacement
0
20
40
60
80
100
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Data
y = 130.00x + 26.30 - R^2 = 0.330

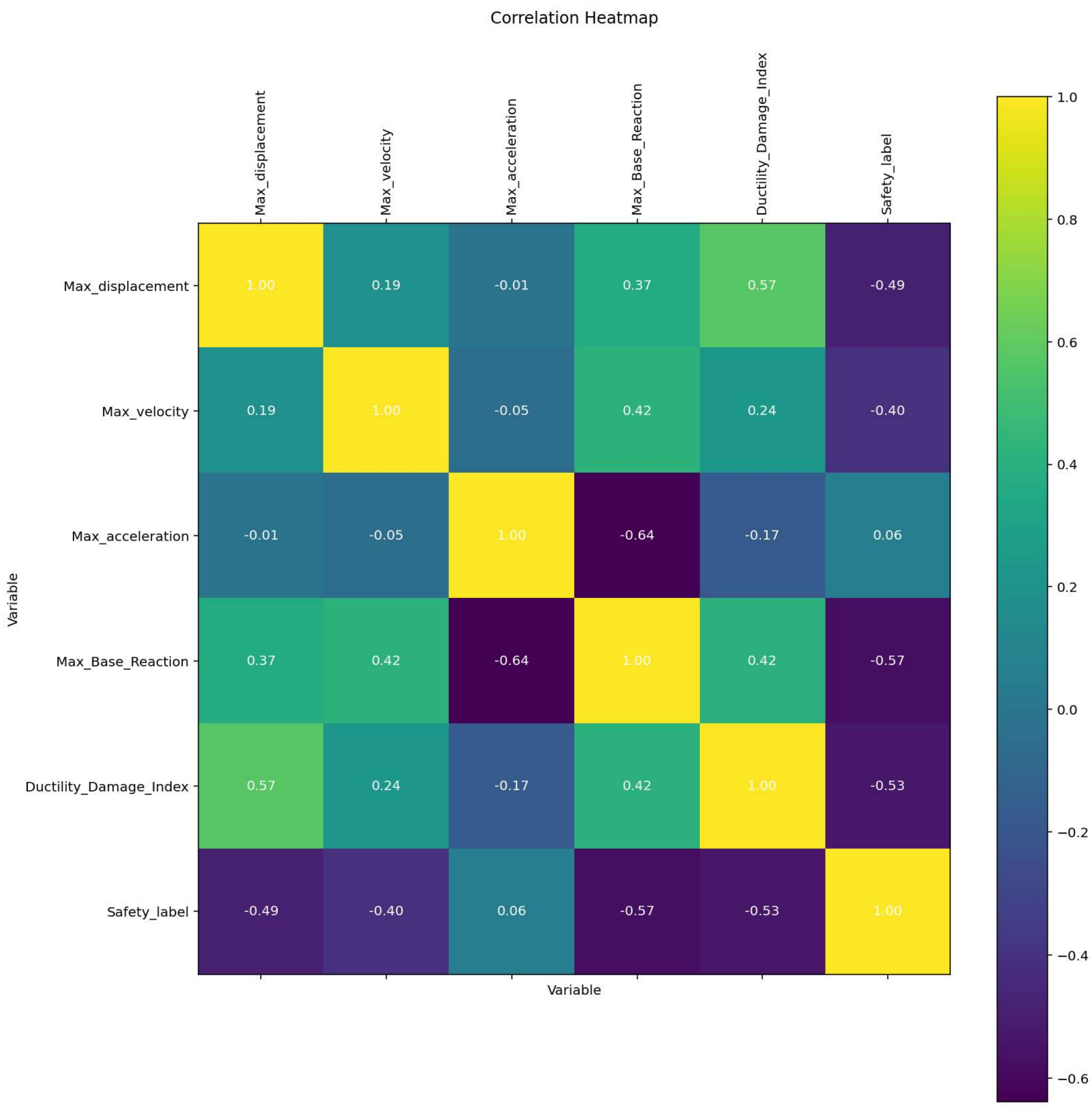


Displacement vs Structural Ductility Damage Index (%) - 3 Clusters (R² = 0.7786)
Cluster 1
Cluster 2
Cluster 3
Structural Ductility Damage Index (%)
Displacement
0
20
40
60
80
100
0.0
0.1
0.2
0.3
0.4
0.5
0.6



Displacement Time History
Displacement [m]
Time [s]
Velocity Time History
Velocity [m/s]
Time [s]
Acceleration Time History
Acceleration [m/s²]
Time [s]
Base Reaction Time History
Base Reaction [N]
Time [s]


Feature Importance
None
Max_displacement
Max_velocity
Max_acceleration
Max_Base_Reaction
Ductility_Damage_Index
0.0
0.1
0.2
0.3
0.4
0.5


Correlation Heatmap
Max_displacement
Max_velocity
Max_acceleration
Max_Base_Reaction
Ductility_Damage_Index
Safety_label
1.00 0.19 -0.01 0.37 0.57 -0.49
0.19 1.00 -0.05 0.42 0.24 -0.40
-0.01 -0.05 1.00 -0.64 -0.17 0.06
0.37 0.42 -0.64 1.00 0.42 -0.57
0.57 0.24 -0.17 0.42 1.00 -0.53
-0.49 -0.40 0.06 -0.57 -0.53 1.00
Variable
Variable
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6